



Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems

MSc Research Internship Report

Alexandre Autran

École Normale Supérieure de Rennes

Supervised by Martin J. Gander
and Bastien Chaudet-Dumas

Mathematics Section
University of Geneva

April – May 2026

Acknowledgments

I warmly thank Martin J. Gander and Bastien Chaudet-Dumas for their guidance, availability, and support throughout this research internship. This project gave me the opportunity to study a stimulating problem and to use the tools in numerical analysis and partial differential equations developed during my MSc year.

Introduction

The Lotka–Volterra system is a classical prey–predator model. Volterra introduced it to describe variations in interacting biological populations [21], and it is now a standard model in mathematical biology [14]. In this report, we study the classical time-dependent equations and their spatial extension with diffusion, using fish populations in a lake as the main interpretation.

The first part presents numerical simulations of the spatial Lotka–Volterra system. The model consists of reaction–diffusion equations with homogeneous Neumann conditions, so the normal population flux satisfies $J_i \cdot n = 0$ on the boundary. We use finite differences, following [8, 15, 16]. The general reaction–diffusion setting is also presented in [10].

The second part introduces an optimal control problem in which fishing acts on the prey population. The cost has the form $J = J_{\text{track}} + J_{\text{effort}} - J_{\text{yield}}$: it measures the distance to target profiles, penalizes the fishing effort, and rewards the harvested quantity. Using optimal control for evolution equations and PDE-constrained problems [12, 20], together with Pontryagin's minimum principle [19], we derive the adjoint equations and characterize the optimal control.

We then discretize the optimality system with a forward–backward sweep algorithm and compare the results with projected-gradient and L-BFGS-B methods. We also study stationary profiles and prove a conditional convergence result for the fully discrete iteration, using the contraction ideas of McAsey, Mou and Han [13].

Contents

1 The Lotka–Volterra System in Time and Space **1**
- 1.1 Numerical Approximation in One Space Dimension . . . 1
- 1.2 Numerical Approximation in Two Space Dimensions . . . 4

2 Control Problems **7**
- 2.1 Model Without a Spatial Variable and Constant Controls . . . 7
- 2.2 Space-time Optimal Control Problem . . . 9
- 2.3 Stationary Profiles . . . 17
- 2.4 Numerical Computation of the Optimal Control . . . 20
- 2.5 Fully Discrete Conditional Convergence of the Forward–Backward Sweep Method . . . 25

1 The Lotka–Volterra System in Time and Space

1.1 Numerical Approximation in One Space Dimension

We start with the classical Lotka–Volterra model [14, 21]. Numerical schemes and qualitative properties are discussed in [3, 7]. Let $v_1(t)$ be the predator density and $v_2(t)$ the prey density at time t. Without spatial dependence, the state is $v(t) = (v_1(t), v_2(t)) \in \mathbf{R}_+^2$. We assume $v_1, v_2 \in \mathscr{C}^1([0,T];\mathbf{R}_+)$, so both densities are differentiable and nonnegative. They solve

$$\begin{cases} \dot{v}_1(t) = v_1(t)\big(-m + \alpha v_2(t)\big), \\ \dot{v}_2(t) = v_2(t)\big(r - \beta v_1(t)\big), \end{cases} \tag{1}$$

The parameters satisfy $m, r, \alpha, \beta > 0$. Here m is the natural predator mortality rate and r is the prey growth rate without predators. The coefficient α measures the positive effect of prey on predator growth, while β measures the loss of prey through predation. To include movement, let $\Omega \subseteq \mathbf{R}^d$ be a bounded habitat and let $x \in \Omega$ denote position. The unknown densities become

$$v_1, v_2 : \Omega \times [0,T] \longrightarrow \mathbf{R}_+.$$

For the pointwise calculations below, we assume

$$v_i \in \mathscr{C}^1([0,T];\mathscr{C}^0(\overline{\Omega})) \cap \mathscr{C}^0([0,T];\mathscr{C}^2(\overline{\Omega})), \qquad i \in \{1,2\}.$$

Under this regularity assumption, every derivative is defined pointwise and the spatial Lotka–Volterra system is

$$\begin{cases} \partial_t v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big) + \mu_1 \Delta v_1(x,t), & \text{in } \Omega \times (0,T), \\ \partial_t v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) + \mu_2 \Delta v_2(x,t), & \text{in } \Omega \times (0,T), \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } \Omega. \end{cases} \tag{2}$$

Here $x \in \mathbf{R}^d$ is the spatial variable, Δ is the Laplace operator, and $\mu_1, \mu_2 > 0$ are the diffusion coefficients. The term $\mu_i \Delta v_i$ models random movement of species i. At the macroscopic scale, Fick's law relates the population flux to the density gradient: $J_i = -\mu_i \nabla v_i$. Thus individuals move, on average, from high-density regions toward low-density regions. If $F_i(v_1, v_2)$ denotes the local Lotka–Volterra reaction, conservation on a small volume gives

$$\partial_t v_i + \nabla \cdot J_i = \text{reaction terms}.$$

After substituting $J_i = -\mu_i \nabla v_i$, this becomes

$$\partial_t v_i - \mu_i \Delta v_i = \text{reaction terms}.$$

The initial densities satisfy $v_1^0, v_2^0 : \Omega \to \mathbf{R}_+$. We impose homogeneous Neumann conditions [6, 10], equivalently $\partial_n v_i = 0$ or $J_i \cdot n = 0$ on $\partial\Omega$. Thus the habitat is closed and no population crosses its boundary. We first take $d = 1$ and $\Omega = (0,1)$, so

$$v_1, v_2 : (0,1) \times [0,T] \longrightarrow \mathbf{R}_+.$$

The Laplacians then reduce to

$$\Delta v_1(x,t) = \partial_x^2 v_1(x,t), \qquad \Delta v_2(x,t) = \partial_x^2 v_2(x,t),$$

and system (2) becomes

$$\begin{cases} \partial_t v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big) + \mu_1 \partial_x^2 v_1(x,t), & \text{in } (0,1) \times (0,T), \\ \partial_t v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) + \mu_2 \partial_x^2 v_2(x,t), & \text{in } (0,1) \times (0,T), \\ \partial_x v_1(0,t) = \partial_x v_1(1,t) = 0, & t \in (0,T), \\ \partial_x v_2(0,t) = \partial_x v_2(1,t) = 0, & t \in (0,T), \\ v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } (0,1). \end{cases} \tag{3}$$

The habitat is the open interval $\Omega = (0,1)$. To include the boundary values, we place the grid on its closure $\overline{\Omega} = [0,1]$: $0 = x_0 < x_1 < \cdots < x_N = 1$. Its spacing is $h = x_{i+1} - x_i = 1/N$, and hence $x_i = ih$ for every $i = 0, \ldots, N$. We represent the two densities by their nodal values

$$v_{1,i}(t) \approx v_1(x_i,t), \qquad v_{2,i}(t) \approx v_2(x_i,t).$$

The semi-discrete state is therefore the pair of time-dependent vectors

$$\mathbf{v}_1(t) = (v_{1,0}(t), \dots, v_{1,N}(t)) \in \mathbf{R}_+^{N+1}, \qquad \mathbf{v}_2(t) = (v_{2,0}(t), \dots, v_{2,N}(t)) \in \mathbf{R}_+^{N+1}.$$



Figure 1: Uniform grid on $\overline{\Omega} = [0, 1]$ for the open habitat $\Omega = (0, 1)$.

At an interior node x_i, $i = 1, \dots, N-1$, Taylor expansion gives the centered finite-difference approximation of order two [8, 16]:

$$\partial_x^2 v_1(x_i, t) \approx \frac{v_{1,i+1}(t) - 2v_{1,i}(t) + v_{1,i-1}(t)}{h^2}, \qquad \partial_x^2 v_2(x_i, t) \approx \frac{v_{2,i+1}(t) - 2v_{2,i}(t) + v_{2,i-1}(t)}{h^2}.$$

The interior semi-discrete equations obtained from system (3) are therefore

$$\begin{cases} \partial_t v_{1,i}(t) = v_{1,i}(t)\big(\alpha v_{2,i}(t) - m\big) + \mu_1 \dfrac{v_{1,i+1}(t) - 2v_{1,i}(t) + v_{1,i-1}(t)}{h^2}, & i = 1, \dots, N-1, \\ \partial_t v_{2,i}(t) = v_{2,i}(t)\big(r - \beta v_{1,i}(t)\big) + \mu_2 \dfrac{v_{2,i+1}(t) - 2v_{2,i}(t) + v_{2,i-1}(t)}{h^2}, & i = 1, \dots, N-1, \\ v_{1,-1} = v_{1,1}, \qquad v_{1,N+1} = v_{1,N-1}, \\ v_{2,-1} = v_{2,1}, \qquad v_{2,N+1} = v_{2,N-1}. \end{cases} \tag{4}$$

At x_0 and x_N, centered differences and ghost points impose the Neumann conditions:

$$\frac{v_{k,1}(t) - v_{k,-1}(t)}{2h} = 0, \qquad \frac{v_{k,N+1}(t) - v_{k,N-1}(t)}{2h} = 0, \qquad k = 1, 2.$$

Hence

$$v_{k,-1}(t) = v_{k,1}(t), \qquad v_{k,N+1}(t) = v_{k,N-1}(t), \qquad k = 1, 2.$$

These ghost values are exactly the discrete form of the zero normal derivative. Substitution into the centered second difference gives the endpoint Laplacians

$$(\delta_{xx} v_k)_0 = \frac{2\big(v_{k,1} - v_{k,0}\big)}{h^2}, \qquad (\delta_{xx} v_k)_N = \frac{2\big(v_{k,N-1} - v_{k,N}\big)}{h^2}, \qquad k = 1, 2.$$

The boundary construction is detailed in [6, 15]. The reference Python code is available in the GitHub repository [2], and a reference simulation video is available here. We choose the localized Gaussian initial densities

$$v_1(x, 0) = 1.5 \exp\left(-\frac{(x - 0.3)^2}{2 \cdot 0.12^2}\right) + 0.02, \qquad v_2(x, 0) = 1.4 \exp\left(-\frac{(x - 0.7)^2}{2 \cdot 0.08^2}\right) + 0.02.$$

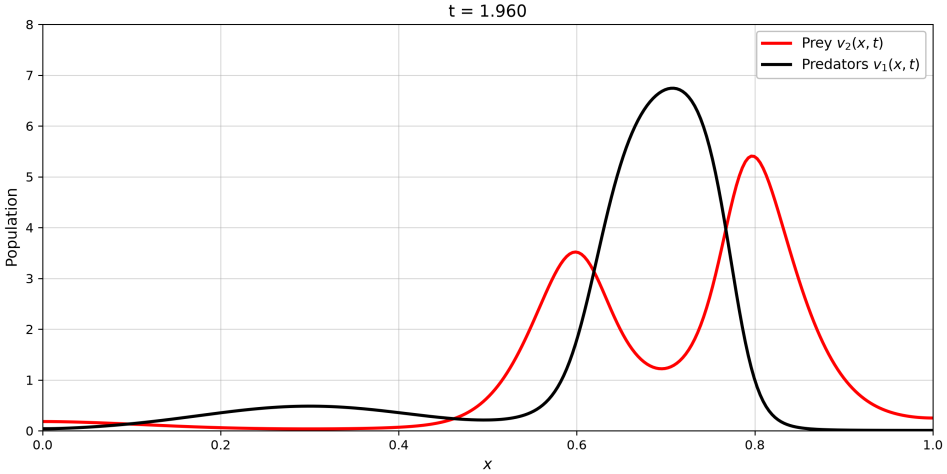


Figure 2: Predator and prey profiles computed with the sequential method at $t = 1.960$, with Gaussian initial data and homogeneous Neumann boundary conditions.

The spatial finite-difference method produces an ODE in $\mathbf{R}^{2(N+1)}$. We use a sequential method of order one in time. We first update prey. We then use the new prey value in the predator reaction. Both diffusion terms use the old values. Figures 2 and 3 are computed with this method, using 300 spatial points, $h = 1/299$, and $\Delta t = 10^{-3}$. We use the uniform time grid

$$0 = t^0 < t^1 < \cdots < t^{N_t} = T, \qquad \Delta t = t^{n+1} - t^n,$$

and write

$$v_{k,i}^n \approx v_k(x_i, t^n), \qquad k = 1, 2.$$

In the reference scripts, u is prey and v is predators. We therefore write $u_i^n = v_{2,i}^n$ and $v_i^n = v_{1,i}^n$. The sequential method gives

$$u_i^{n+1} = u_i^n + \Delta t \left[u_i^n (b - \beta v_i^n) + \nu_u \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{h^2} \right],$$

and

$$v_i^{n+1} = v_i^n + \Delta t \left[v_i^n (\alpha u_i^{n+1} - a) + \nu_v \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{h^2} \right],$$

for $i = 1, \ldots, N-1$. Here $a = m$, $b = r$, $\nu_u = \mu_2$, and $\nu_v = \mu_1$. For the one-dimensional plots, we set $m = 0.6$, $r = 1.2$, $\alpha = \beta = 1$, $\mu_1 = 10^{-4}$, and $\mu_2 = 2 \times 10^{-4}$. For Neumann conditions, we use the ghost-point Laplacian at $i = 0$ and $i = N$. The method has order two in space at interior nodes and order one in time. Without diffusion, the reaction step is $U = u[1 + \Delta t(b - \beta v)]$ and $V = v[1 + \Delta t(\alpha U - a)]$. Its Jacobian determinant is $UV/(uv)$. Thus it preserves the Lotka–Volterra Poisson tensor $uv \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ when $u, v, U, V > 0$. The reaction step is therefore a Poisson integrator. For a smooth solution and an inactive positivity projection, the formal error is $O(\Delta t + h^2)$. Stability imposes an additional relation between h and Δt. If only diffusion is retained, a sufficient condition is

$$\Delta t \leqslant \frac{h^2}{2 \max(\nu_u, \nu_v)},$$

The reaction terms also enter the positivity bounds. If the old values are nonnegative, the following bounds ensure that the new values are nonnegative:

$$\Delta t \left(\beta \max_i v_i^n + \frac{2\nu_u}{h^2} \right) \leqslant 1, \qquad \Delta t \left(a + \frac{2\nu_v}{h^2} \right) \leqslant 1.$$

Under these bounds, all coefficients in the prey update are nonnegative. Hence $u^{n+1} \geqslant 0$, and all coefficients in the predator update are also nonnegative. The same argument applies at the boundary. The code [2] replaces negative values by zero using $z \mapsto \max(z, 0)$. Under the bounds above, the update already preserves nonnegativity and this projection is inactive [3, 7].

To compare boundary effects, we repeat the one-dimensional experiment with homogeneous Dirichlet conditions,

$$v_1(0, t) = v_1(1, t) = 0, \qquad v_2(0, t) = v_2(1, t) = 0.$$

The discrete boundary values are then fixed by

$$v_{1,0}^n = v_{1,N}^n = 0, \qquad v_{2,0}^n = v_{2,N}^n = 0, \qquad n = 0, \ldots, N_t.$$

Figure 3 shows the resulting profiles.

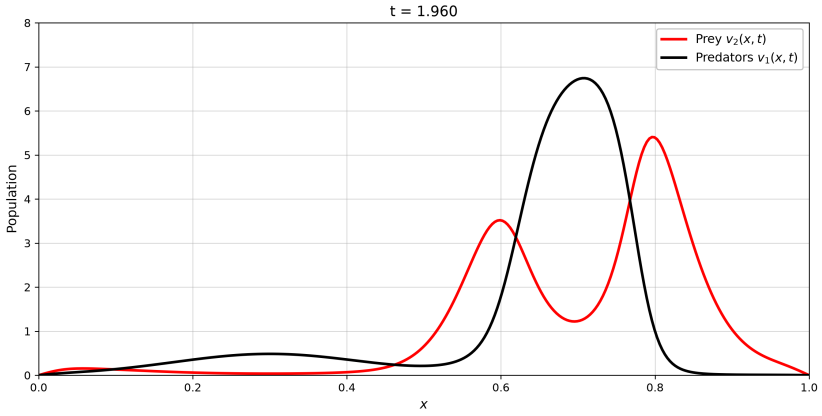


Figure 3: Predator and prey profiles computed with the sequential method at $t = 1.960$, with Gaussian initial data and homogeneous Dirichlet boundary conditions.

Remark. Gaussian data describe populations initially concentrated in small regions. Diffusion then spreads each profile. Constant data, sinusoidal perturbations, step functions, and sums of Gaussians produce different transients. Neumann conditions model a closed habitat, while Dirichlet conditions model an absorbing or hostile boundary.

1.2 Numerical Approximation in Two Space Dimensions

The same finite-difference construction extends to $d = 2$. We take $(x, y) \in \mathbf{R}^2$ and use the framework of [8, 16]. The densities are now

$$v_1, v_2 : (0,1)^2 \times [0,T] \longrightarrow \mathbf{R}_+.$$

For $w = w(x, y, t)$, the two-dimensional Laplacian is $\Delta w = \partial_x^2 w + \partial_y^2 w$. Hence

$$\Delta v_1(x,y,t) = \partial_x^2 v_1(x,y,t) + \partial_y^2 v_1(x,y,t), \qquad \Delta v_2(x,y,t) = \partial_x^2 v_2(x,y,t) + \partial_y^2 v_2(x,y,t).$$

System (2) becomes

$$\begin{cases} \partial_t v_1(x,y,t) = v_1(x,y,t)\big(\alpha v_2(x,y,t) - m\big) + \mu_1 \Delta v_1(x,y,t), & \text{in } (0,1)^2 \times (0,T), \\ \partial_t v_2(x,y,t) = v_2(x,y,t)\big(r - \beta v_1(x,y,t)\big) + \mu_2 \Delta v_2(x,y,t), & \text{in } (0,1)^2 \times (0,T), \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial(0,1)^2 \times (0,T), \\ v_1(\cdot,\cdot,0) = v_1^0, \qquad v_2(\cdot,\cdot,0) = v_2^0, & \text{in } (0,1)^2. \end{cases}$$

The two-dimensional habitat is the open square $\Omega = (0,1)^2$. We place a uniform grid on its closure $\overline{\Omega} = [0,1]^2$, as shown in Figure 4:

$$0 = x_0 < \cdots < x_{N_x} = 1, \qquad 0 = y_0 < \cdots < y_{N_y} = 1,$$

where $h_x = N_x^{-1}$, $h_y = N_y^{-1}$, $x_i = ih_x$, and $y_j = jh_y$. On the square, it is natural to choose $N_x = N_y = N$, so $h_x = h_y = 1/N$. We keep h_x and h_y separate in the formulas to allow different spacings in the two directions. The nodal values satisfy

$$v_{1,i,j}(t) \approx v_1(x_i, y_j, t), \qquad v_{2,i,j}(t) \approx v_2(x_i, y_j, t).$$

The semi-discrete state consists of the arrays

$$\mathbf{v}_1(t) = \big(v_{1,i,j}(t)\big)_{0 \leqslant i \leqslant N_x,\, 0 \leqslant j \leqslant N_y}, \qquad \mathbf{v}_2(t) = \big(v_{2,i,j}(t)\big)_{0 \leqslant i \leqslant N_x,\, 0 \leqslant j \leqslant N_y},$$

so $(\mathbf{v}_1(t), \mathbf{v}_2(t)) \in \big(\mathbf{R}_+^{(N_x+1)\times(N_y+1)}\big)^2$.

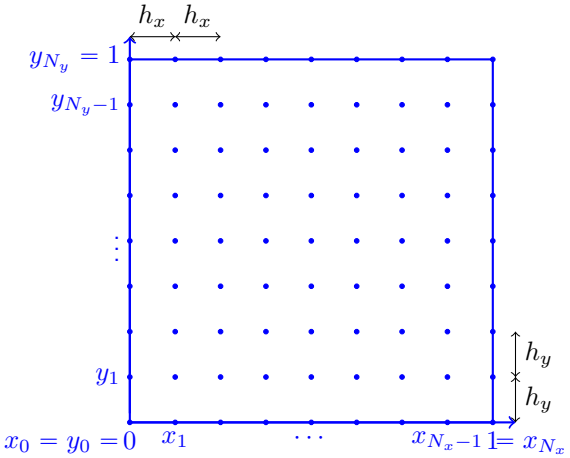


Figure 4: Uniform grid on $\overline{\Omega} = [0,1]^2$ for the open habitat $\Omega = (0,1)^2$.

At an interior node (x_i, y_j), centered second differences give [8, 15, 16]

$$\partial_x^2 v_1(x_i,y_j,t) \approx \frac{v_{1,i+1,j}(t) - 2v_{1,i,j}(t) + v_{1,i-1,j}(t)}{h_x^2}, \qquad \partial_y^2 v_1(x_i,y_j,t) \approx \frac{v_{1,i,j+1}(t) - 2v_{1,i,j}(t) + v_{1,i,j-1}(t)}{h_y^2},$$

and therefore

$$\Delta v_1(x_i, y_j, t) \approx \frac{v_{1,i+1,j}(t) - 2v_{1,i,j}(t) + v_{1,i-1,j}(t)}{h_x^2} + \frac{v_{1,i,j+1}(t) - 2v_{1,i,j}(t) + v_{1,i,j-1}(t)}{h_y^2}.$$

The same five-point stencil for v_2 gives

$$\Delta v_2(x_i, y_j, t) \approx \frac{v_{2,i+1,j}(t) - 2v_{2,i,j}(t) + v_{2,i-1,j}(t)}{h_x^2} + \frac{v_{2,i,j+1}(t) - 2v_{2,i,j}(t) + v_{2,i,j-1}(t)}{h_y^2}.$$

The interior semi-discrete system is

$$\begin{cases} \partial_t v_{1,i,j} = v_{1,i,j}\big(\alpha v_{2,i,j} - m\big) + \mu_1 \left(\dfrac{v_{1,i+1,j} - 2v_{1,i,j} + v_{1,i-1,j}}{h_x^2} + \dfrac{v_{1,i,j+1} - 2v_{1,i,j} + v_{1,i,j-1}}{h_y^2} \right), & \substack{i=1,\dots,N_x-1,\\ j=1,\dots,N_y-1,} \\ \partial_t v_{2,i,j} = v_{2,i,j}\big(r - \beta v_{1,i,j}\big) + \mu_2 \left(\dfrac{v_{2,i+1,j} - 2v_{2,i,j} + v_{2,i-1,j}}{h_x^2} + \dfrac{v_{2,i,j+1} - 2v_{2,i,j} + v_{2,i,j-1}}{h_y^2} \right), & \substack{i=1,\dots,N_x-1,\\ j=1,\dots,N_y-1,} \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, \qquad \text{on the boundary nodes.} \end{cases}$$

The boundary stencil depends on the boundary condition. For homogeneous Neumann conditions, we use the same reflected ghost-point construction as in (4). If $w_{i,j}$ denotes either v_1 or v_2, then $\partial_n w = 0$ is approximated by centered first differences. This gives

$$w_{-1,j} = w_{1,j}, \qquad w_{N_x+1,j} = w_{N_x-1,j}$$

on the vertical sides, and

$$w_{i,-1} = w_{i,1}, \qquad w_{i,N_y+1} = w_{i,N_y-1}$$

on the horizontal sides. Inserting the reflected values into the centered five-point stencil gives the boundary version of Δ_h. For example, at a node on the left side, $i = 0$ and $1 \leqslant j \leqslant N_y - 1$, we obtain

$$(\Delta_h w)_{0,j} = \frac{2(w_{1,j} - w_{0,j})}{h_x^2} + \frac{w_{0,j+1} - 2w_{0,j} + w_{0,j-1}}{h_y^2}.$$

The other three sides are obtained in the same way.

In two dimensions, we again update prey before predators. The two-dimensional plots use this sequential method with 80 points in each direction, $h_x = h_y = 1/79$, and $\Delta t = 5 \times 10^{-4}$. We set $m = 0.6$, $r = 1.2$, $\alpha = \beta = 1$, $\mu_1 = 5 \times 10^{-5}$, and $\mu_2 = 8 \times 10^{-5}$. We use the time grid

$$0 = t^0 < t^1 < \cdots < t^{N_t} = T, \qquad \Delta t = t^{n+1} - t^n,$$

and denote

$$u_{i,j}^n \approx v_2(x_i, y_j, t^n), \qquad v_{i,j}^n \approx v_1(x_i, y_j, t^n).$$

As in one dimension, $u_{i,j}^n$ approximates prey and $v_{i,j}^n$ approximates predators. We first compute u^{n+1}. We then use it in the predator reaction:

$$u_{i,j}^{n+1} = u_{i,j}^n + \Delta t \left[u_{i,j}^n (b - \beta v_{i,j}^n) + \nu_u \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h_x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h_y^2} \right) \right],$$

and

$$v_{i,j}^{n+1} = v_{i,j}^n + \Delta t \left[v_{i,j}^n (\alpha u_{i,j}^{n+1} - a) + \nu_v \left(\frac{v_{i+1,j}^n - 2v_{i,j}^n + v_{i-1,j}^n}{h_x^2} + \frac{v_{i,j+1}^n - 2v_{i,j}^n + v_{i,j-1}^n}{h_y^2} \right) \right],$$

at the interior nodes. The reference scripts use 80 points in each direction, so $h_x = h_y = 1/79$. In the code, `Nx` and `Ny` count points. Here N_x and N_y count intervals. For Neumann conditions, we use the reflected ghost values at boundary nodes. The method has order one in time. The five-point Laplacian has order two in each spatial direction. Without diffusion or projection, the reaction step at each node preserves the same Poisson tensor as in one dimension. Thus, when the positivity projection is inactive and the exact solution is smooth, the formal error has the form $O(\Delta t + h_x^2 + h_y^2)$. Stability still requires a sufficiently small step. For the diffusion part alone, a sufficient condition is

$$\Delta t \, \max(\nu_u, \nu_v) \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) \leqslant \frac{1}{2},$$

The reaction terms also enter the positivity bounds. Set $\sigma_h = h_x^{-2} + h_y^{-2}$. For nonnegative old values, the following bounds ensure nonnegative new values:

$$\Delta t \left(\beta \max_{i,j} v_{i,j}^n + 2\nu_u \sigma_h \right) \leqslant 1, \qquad \Delta t \, (a + 2\nu_v \sigma_h) \leqslant 1.$$

The proof is the same as in one dimension: all update coefficients are nonnegative. These bounds also work for explicit Euler, which uses the old prey value in the predator reaction. The code [2] replaces negative values by zero. Under the bounds above, the update preserves nonnegativity and this projection is inactive.

The plots below use the sequential method above, with the parameters and initial profiles from the reference Python code [2]. A reference video is available here. The initial densities are localized Gaussian functions:

$$v_1(x, y, 0) = 1.3 \, \exp\left(- \left(\frac{(x - 0.35)^2}{2 \cdot 0.12^2} + \frac{(y - 0.42)^2}{2 \cdot 0.10^2} \right) \right) + 0.02,$$

$$v_2(x, y, 0) = 1.4 \, \exp\left(- \left(\frac{(x - 0.68)^2}{2 \cdot 0.10^2} + \frac{(y - 0.60)^2}{2 \cdot 0.12^2} \right) \right) + 0.02.$$

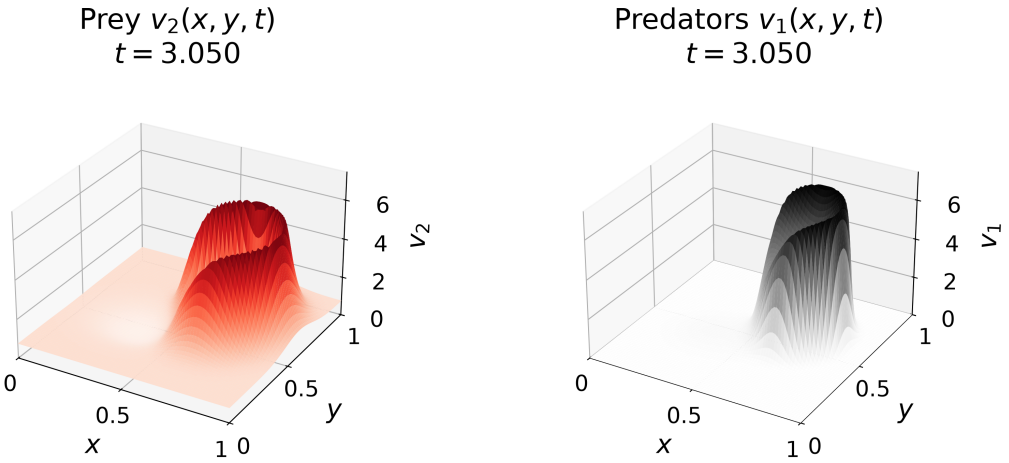


Figure 5: Solution profiles computed with the sequential method at $t = 3.050$, with homogeneous Neumann boundary conditions and localized Gaussian initial data.

For comparison, the next figure uses homogeneous Dirichlet conditions. We keep the same interior initial values and set every boundary value to zero from the initial step.

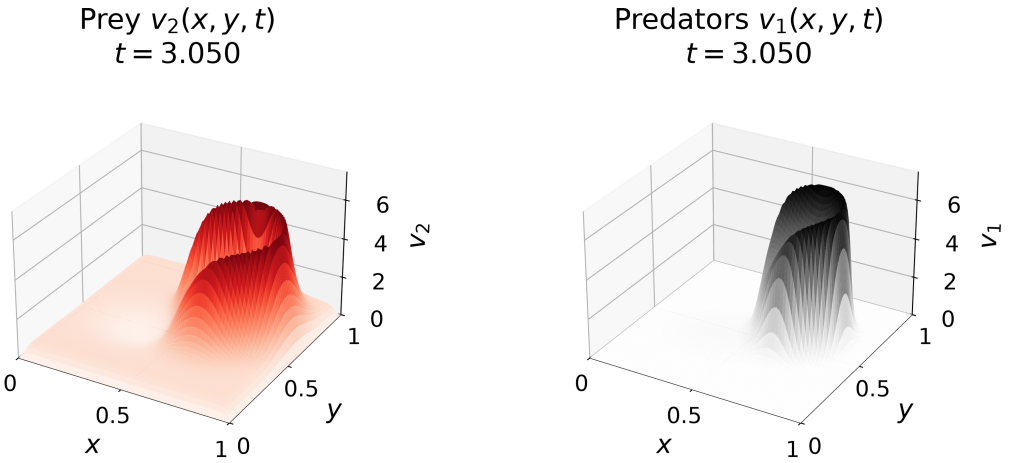


Figure 6: Solution profiles computed with the sequential method at $t = 3.050$, with homogeneous Dirichlet boundary conditions and localized Gaussian initial data. Figures 5 and 6 use the same time and the same vertical and color scales.

2 Control Problems

We now pass from the simulation of uncontrolled dynamics to the design of a fishing policy. We first study the non-spatial system with a constant effort $u_2(t) = \bar{u}_2$, compute all equilibria, and determine their local stability. This calculation gives the threshold $q\bar{u}_2 = r$. We then allow u_2 to depend on (x, t), formulate a distributed optimal control problem, derive the state and adjoint equations, and approximate the optimality conditions numerically.

2.1 Model Without a Spatial Variable and Constant Controls

The controlled time-dependent model is the natural extension of the classical prey–predator system [14, 21]:

$$\begin{cases} \dot{v}_1(t) = v_1(t)\big(-m + \alpha v_2(t)\big) \\ \dot{v}_2(t) = v_2(t)\big(r - \beta v_1(t)\big) - q\, u_2(t)\, v_2(t) \end{cases}$$

with initial values $v_1(0) = v_1^0 \geqslant 0$ and $v_2(0) = v_2^0 \geqslant 0$. All parameters are positive. The coefficient m is the natural mortality rate of predators, α measures the increase of predator growth produced by prey, r is the prey growth rate without predators, β measures predation losses, and q converts the fishing effort u_2 into an additional prey mortality qu_2. Thus the controlled prey per-capita growth rate is $r - \beta v_1 - qu_2$. Assume $u_2(t) \geqslant 0$ and $(v_1^0, v_2^0) \in \mathbf{R}_+^2$. Since each right-hand side vanishes on its corresponding coordinate axis, the positive quadrant $\mathbf{R}_+^2$ is invariant. We first set $u_2(t) = \bar{u}_2 \geqslant 0$. An equilibrium (v_1^*, v_2^*) satisfies

$$\begin{cases} 0 = v_1^*\big(-m + \alpha v_2^*\big) \\ 0 = v_2^*\big(r - \beta v_1^*\big) - q\bar{u}_2\, v_2^* \end{cases}.$$

The extinction state $E_0 = (0, 0)$ is always an equilibrium. If $v_1^* = 0$ and $v_2^* > 0$, the second equation gives $v_2^*(r - q\bar{u}_2) = 0$. Hence a positive predator-free equilibrium exists only in the critical case $r - q\bar{u}_2 = 0$. In that case, every point of the half-line $\{(0, s) : s \geqslant 0\}$ is stationary because prey growth and fishing mortality cancel exactly. If $r - q\bar{u}_2 \neq 0$, there is no positive predator-free equilibrium. For coexistence, both components are positive, so the two factors multiplying v_1^* and v_2^* must vanish:

$$-m + \alpha v_2^* = 0, \qquad r - \beta v_1^* - q\bar{u}_2 = 0,$$

which gives

$$v_2^* = \frac{m}{\alpha}, \qquad v_1^* = \frac{r - q\bar{u}_2}{\beta}.$$

Hence a positive coexistence equilibrium exists if and only if $r - q\bar{u}_2 > 0$, and then

$$E_* = \left(\frac{r - q\bar{u}_2}{\beta}, \frac{m}{\alpha}\right).$$

Remark. The coexistence condition is $q\bar{u}_2 < r$. When $q\bar{u}_2 \geqslant r$, fishing removes the positive coexistence equilibrium because prey growth no longer sustains the predator population.

Set $\rho := r - q\bar{u}_2$. The controlled system becomes

$$\begin{cases} \dot{v}_1 = v_1(-m + \alpha v_2) \\ \dot{v}_2 = v_2(\rho - \beta v_1) \end{cases}.$$

Assume first that $\rho = r - q\bar{u}_2 > 0$. The positive equilibrium is $E_* = (\rho/\beta, m/\alpha)$. The classical Lotka–Volterra Hamiltonian provides a direct stability argument [14]. Define

$$H(v_1, v_2) = \beta v_1 - \rho \ln v_1 + \alpha v_2 - m \ln v_2, \qquad \text{with } v_1 > 0,\ v_2 > 0.$$

For every trajectory in $(0, +\infty)^2$,

$$\frac{\mathrm{d}}{\mathrm{d}t} H(v_1(t), v_2(t)) = 0.$$

Indeed,

$$\frac{\partial H}{\partial v_1} = \beta - \frac{\rho}{v_1}, \qquad \frac{\partial H}{\partial v_2} = \alpha - \frac{m}{v_2}.$$

The chain rule gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}H(v_1(t),v_2(t)) &= \left(\beta-\frac{\rho}{v_1}\right)\dot{v}_1+\left(\alpha-\frac{m}{v_2}\right)\dot{v}_2\\ &= \left(\beta-\frac{\rho}{v_1}\right)v_1(-m+\alpha v_2)+\left(\alpha-\frac{m}{v_2}\right)v_2(\rho-\beta v_1)\\ &= (\beta v_1-\rho)(-m+\alpha v_2)+(\alpha v_2-m)(\rho-\beta v_1)\\ &= 0.\end{aligned}$$

Thus $H(v_1(t),v_2(t)) = H(v_1(0),v_2(0))$. Moreover, E_* is a strict local minimum of H, because

$$\frac{\partial H}{\partial v_1}(E_*)=0, \qquad \frac{\partial H}{\partial v_2}(E_*)=0,$$

and

$$\frac{\partial^2 H}{\partial v_1^2}(E_*)=\frac{\rho}{(v_1^*)^2}>0, \qquad \frac{\partial^2 H}{\partial v_2^2}(E_*)=\frac{m}{(v_2^*)^2}>0, \qquad \frac{\partial^2 H}{\partial v_1\partial v_2}=0.$$

The nearby level sets $\{H=c\}$ are closed curves around E_*. A trajectory stays on one such curve, so nearby solutions remain nearby and oscillate around the equilibrium. Therefore E_* is a center: it is Lyapunov stable and not asymptotically stable.

At the extinction state, the Jacobian is

$$J(E_0)=\begin{pmatrix}-m & 0\\ 0 & r-q\bar{u}_2\end{pmatrix}.$$

Its eigenvalues are $-m$ and $r-q\bar{u}_2$. Together with the equilibrium calculation, this gives the three regimes

- $r-q\bar{u}_2<0$: $E_0=(0,0)$ is locally asymptotically stable,
- $r-q\bar{u}_2=0$: the equilibrium set is $\{(0,s): s\geqslant 0\}$,
- $r-q\bar{u}_2>0$: $E_*=((r-q\bar{u}_2)/\beta, m/\alpha)$ exists and is a center, so it is locally stable and not asymptotically stable.

In the critical case $r-q\bar{u}_2=0$, natural prey growth and fishing mortality balance exactly when $v_1=0$. Indeed, the prey equation reduces to

$$\dot{v}_2=(r-q\bar{u}_2)v_2.$$

Thus $r-q\bar{u}_2=0$ gives $\dot{v}_2=0$ in the absence of predators. If $v_1>0$, the prey equation becomes

$$\dot{v}_2=-\beta v_1 v_2,$$

and therefore $\dot{v}_2<0$ whenever $v_1v_2>0$.

The Python experiment [2] illustrates the three regimes with the same initial condition. We select three constant efforts $\bar{u}_2$: one with $r-q\bar{u}_2<0$, one with $r-q\bar{u}_2=0$, and one with $r-q\bar{u}_2>0$. The resulting trajectories show extinction, the critical balance, and closed coexistence oscillations.

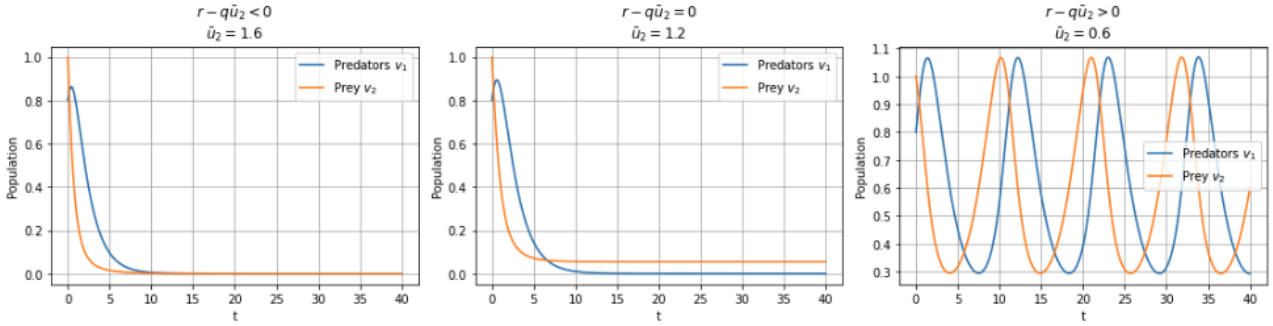
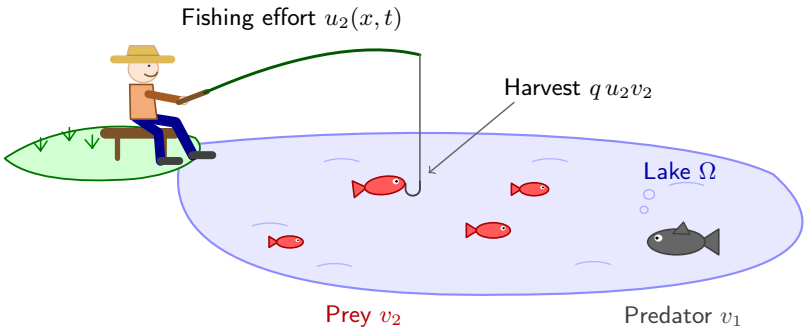


Figure 7: The three regimes of the controlled Lotka–Volterra ODE: extinction for $r-q\bar{u}_2<0$, critical predator-free balance for $r-q\bar{u}_2=0$, and coexistence oscillations for $r-q\bar{u}_2>0$.

2.2 Space-time Optimal Control Problem

We now let the fishing effort depend on both variables, $u_2 = u_2(x,t)$. The formulation follows optimal control for evolution equations [12], PDE-constrained optimization [20], and Pontryagin's principle [19]. The harvested prey quantity is $\int_Q q\,u_2 v_2\,\mathrm{d}x\,\mathrm{d}t$, so maximizing yield is equivalent to minimizing its negative. The state is defined on $Q = \Omega \times (0,T)$, where $\Omega \subseteq \mathbf{R}^d$ is bounded. Both populations diffuse, with $\mu_1, \mu_2 > 0$, and fishing adds the mortality term $-qu_2v_2$ to the prey equation. The controlled system is

$$\begin{cases} \partial_t v_1(x,t) - \mu_1 \Delta v_1(x,t) = v_1(x,t)\big(-m + \alpha v_2(x,t)\big), & \text{in } \Omega \times (0,T), \\ \partial_t v_2(x,t) - \mu_2 \Delta v_2(x,t) = v_2(x,t)\big(r - \beta v_1(x,t)\big) - q\,u_2(x,t)\,v_2(x,t), & \text{in } \Omega \times (0,T), \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } \Omega. \end{cases}$$



The fisher represents the control $u_2(x,t)$. Fishing removes prey from the lake at rate $q\,u_2(x,t)v_2(x,t)$.

The value $u_2(x,t)$ measures the local fishing effort exerted on prey at $(x,t) \in Q$. We prescribe target densities $v_1^{\mathrm{tar}}, v_2^{\mathrm{tar}} \in L^2(Q)$. Targets are functions of space and time. A stationary target is independent of t. Physical and operational limits are represented by a pointwise upper bound $U_2^{\max} > 0$. The admissible control set is

$$\mathscr{U}_{\mathrm{ad}}^{\Omega} := \{u_2 \in L^\infty(Q) \;\mid\; 0 \leqslant u_2(x,t) \leqslant U_2^{\max} \text{ a.e. in } Q\}\,.$$

For $(v_1, v_2, u_2) \in L^2(Q)^3$, define

$$\mathscr{J} : L^2(Q) \times L^2(Q) \times L^2(Q) \longrightarrow \mathbf{R}$$

by

$$\begin{aligned} \mathscr{J}(v_1,v_2,u_2) := \frac{1}{2}\int_0^T \int_\Omega \Big[\gamma_1\big(v_1(x,t) - v_1^{\mathrm{tar}}(x,t)\big)^2 + \gamma_2\big(v_2(x,t) - v_2^{\mathrm{tar}}(x,t)\big)^2 + \lambda_2 u_2(x,t)^2\Big]\,\mathrm{d}x\,\mathrm{d}t \\ - \eta \int_0^T \int_\Omega q\,u_2(x,t)\,v_2(x,t)\,\mathrm{d}x\,\mathrm{d}t. \end{aligned}$$

Let $S(u_2) = (v_1, v_2)$ be the state generated by u_2. The reduced functional is

$$J : \mathscr{U}_{\mathrm{ad}}^{\Omega} \longrightarrow \mathbf{R}, \qquad J(u_2) := \mathscr{J}(S(u_2), u_2).$$

The three terms have the following roles:

- $\gamma_i > 0$ weights the tracking error $\|v_i - v_i^{\mathrm{tar}}\|^2_{L^2(Q)}$, $i = 1, 2$,
- $\lambda_2 > 0$ weights the effort $\|u_2\|^2_{L^2(Q)}$,
- $-\eta q \int_Q u_2 v_2\,\mathrm{d}x\,\mathrm{d}t$ rewards the harvested quantity.

Remark. Minimizing $J = J_{\mathrm{track}} + J_{\mathrm{effort}} - J_{\mathrm{yield}}$ balances population targets, fishing cost, and yield.

Before minimizing J, we verify that each admissible control produces one state on the whole interval $[0,T]$. We work with weak solutions in the standard parabolic setting [6]. The semilinear theory and the qualitative properties used below are also developed in [9, 18]. A bounded Lipschitz domain is sufficient. The initial values are understood in

the trace sense, and the variational formulation incorporates the homogeneous Neumann condition. Set $V = H^1(\Omega)$ and let V' be its dual. For a fixed $u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$, we seek states with the regularity

$$v_i \in L^2(0,T;H^1(\Omega)) \cap L^\infty(Q), \qquad \partial_t v_i \in L^2(0,T;V'), \qquad i = 1,2.$$

The equations hold in $L^2(0,T;V')$. For $i \in \{1,2\}$, the diffusion term and the weak Neumann condition are encoded by

$$\langle \partial_t v_i, \varphi \rangle_{V',V} + \mu_i \int_\Omega \nabla v_i \cdot \nabla \varphi \, \mathrm{d}x = \int_\Omega F_i \, \varphi \, \mathrm{d}x, \qquad \forall \varphi \in H^1(\Omega),$$

for a.e. $t \in (0,T)$, where F_i is the reaction term of species i.

Proposition 2.1 (Well-posed state system)

Let $\Omega \subseteq \mathbf{R}^d$ be bounded and Lipschitz, set $Q = \Omega \times (0,T)$, and assume $\mu_1, \mu_2, m, \alpha, r, \beta, q > 0$. Let

$$u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega} = \{u_2 \in L^\infty(Q) \ \mid \ 0 \leqslant u_2(x,t) \leqslant U_2^{\max} \text{ a.e. in } Q\}.$$

Assume $v_1^0, v_2^0 \in L^\infty(\Omega)$ and $v_1^0, v_2^0 \geqslant 0$ a.e. in Ω. Then the system

$$\begin{cases} \partial_t v_1 - \mu_1 \Delta v_1 = v_1(-m + \alpha v_2), & \text{in } Q, \\ \partial_t v_2 - \mu_2 \Delta v_2 = v_2(r - \beta v_1) - q u_2 v_2, & \text{in } Q, \\ \dfrac{\partial v_1}{\partial n} = \dfrac{\partial v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ v_1(\cdot,0) = v_1^0, \qquad v_2(\cdot,0) = v_2^0, & \text{in } \Omega, \end{cases}$$

has a unique bounded weak solution on every finite interval $[0,T]$, with

$$v_i \in L^2(0,T;H^1(\Omega)) \cap L^\infty(Q), \qquad \partial_t v_i \in L^2(0,T;V'), \qquad i = 1,2.$$

Moreover, $(v_1, v_2) \in L^\infty(Q; \mathbf{R}^2_+)$. This means that $v_1, v_2 \geqslant 0$ a.e. in Q.

Proof. Standard semilinear parabolic theory provides local existence, uniqueness on bounded sets, and a continuation criterion [1, 9]. The weak comparison principle used for nonnegativity is recalled in [18]. Fix $u_2 \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$. Extend u_2 by zero for $t > T$. To apply these results, write the system in the form

$$\partial_t v_1 - \mu_1 \Delta v_1 = f_1(v_1, v_2), \qquad \partial_t v_2 - \mu_2 \Delta v_2 = f_2(x,t,v_1,v_2),$$

where

$$\begin{cases} f_1(v_1,v_2) := v_1(-m + \alpha v_2) \\ f_2(x,t,v_1,v_2) := v_2(r - \beta v_1) - q u_2(x,t) v_2. \end{cases}$$

Because $u_2 \in L^\infty(Q)$, f_1 and f_2 are locally Lipschitz in (v_1, v_2) on every bounded subset of $\mathbb{R}^2$, uniformly in (x,t). The local existence theorem for semilinear parabolic systems with Neumann conditions [1] gives a maximal weak solution on $[0, T_{\max})$ such that

$$v_i \in L^2(0,\tau;H^1(\Omega)) \cap L^\infty(\Omega \times (0,\tau)), \qquad \partial_t v_i \in L^2(0,\tau;V'),$$

for every $\tau < T_{\max}$ and $i = 1,2$. We next prove positivity. Since $f_1(0,v_2) = 0$ and $f_2(x,t,v_1,0) = 0$, both coordinate axes are invariant. Equivalently, test the equations with the negative parts

$$v_1^- := \max(-v_1, 0), \qquad v_2^- := \max(-v_2, 0),$$

The Neumann boundary terms vanish, and the resulting inequalities control v_1^- and v_2^-. Since $v_1^0, v_2^0 \geqslant 0$, the weak maximum principle [6, 18] gives $(v_1, v_2) \geqslant (0,0)$ a.e. on $\Omega \times (0, T_{\max})$. We now obtain L^∞ bounds. From $v_1, u_2 \geqslant 0$,

$$r - \beta v_1 - q u_2 \leqslant r.$$

Thus

$$\partial_t v_2 - \mu_2 \Delta v_2 = v_2(r - \beta v_1 - q u_2) \leqslant r v_2.$$

Compare v_2 with $z' = rz$, $z(0) = \|v_2^0\|_{L^\infty(\Omega)}$. For $0 < t < T_{\max}$ this gives

$$0 \leqslant v_2(x,t) \leqslant \|v_2^0\|_{L^\infty(\Omega)} e^{rt} \quad \text{for a.e. } x \in \Omega.$$

Hence, for each $\tau < T_{\max}$,

$$\|v_2\|_{L^\infty(\Omega\times(0,\tau))} \leqslant \|v_2^0\|_{L^\infty(\Omega)} e^{r\tau}.$$

Define $M_2(\tau) := \|v_2^0\|_{L^\infty(\Omega)} e^{r\tau}$. Then

$$0 \leqslant v_2(x,t) \leqslant M_2(\tau) \quad \text{for a.e. } (x,t) \in \Omega \times (0,\tau).$$

The first state equation now gives

$$\partial_t v_1 - \mu_1 \Delta v_1 = v_1(-m + \alpha v_2) \leqslant \alpha M_2(\tau) v_1 \quad \text{in } \Omega \times (0,\tau).$$

Compare this inequality with $y' = \alpha M_2(\tau) y$, $y(0) = \|v_1^0\|_{L^\infty(\Omega)}$. We obtain

$$0 \leqslant v_1(x,t) \leqslant \|v_1^0\|_{L^\infty(\Omega)} e^{\alpha M_2(\tau) t} \quad \text{for a.e. } (x,t) \in \Omega \times (0,\tau).$$

Therefore

$$\|v_1\|_{L^\infty(\Omega\times(0,\tau))} \leqslant \|v_1^0\|_{L^\infty(\Omega)} e^{\alpha M_2(\tau)\tau}.$$

These estimates prevent finite-time blow-up in L^∞. Indeed, if $T_{\max} < +\infty$, the continuation criterion for semilinear parabolic systems would require

$$\limsup_{t \to T_{\max}^-} \left(\|v_1(t)\|_{L^\infty(\Omega)} + \|v_2(t)\|_{L^\infty(\Omega)} \right) = +\infty.$$

The estimates above exclude this blow-up alternative, so $T_{\max} = +\infty$ and the solution exists on the prescribed interval $[0,T]$. It remains to prove uniqueness. Let (v_1, v_2) and (w_1, w_2) be two bounded weak solutions associated with the same control and the same initial data, and set $z_1 = v_1 - w_1$, $z_2 = v_2 - w_2$. Both solutions remain in a common bounded subset of $\mathbb{R}^2$ for a.e. $(x,t) \in Q$. On this set, the nonlinearities f_1 and f_2 have a common Lipschitz constant. Consequently, there is $L_T > 0$ such that, for a.e. $t \in (0,T)$,

$$\|f_1(v_1,v_2) - f_1(w_1,w_2)\|_{L^2(\Omega)} + \|f_2(\cdot,t,v_1,v_2) - f_2(\cdot,t,w_1,w_2)\|_{L^2(\Omega)} \leqslant L_T \left(\|z_1(t)\|_{L^2(\Omega)} + \|z_2(t)\|_{L^2(\Omega)} \right).$$

Subtract the two systems, test the equations with z_1 and z_2, integrate over Ω, and use the weak Neumann conditions. This gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right) + \mu_1 \|\nabla z_1(t)\|_{L^2(\Omega)}^2 + \mu_2 \|\nabla z_2(t)\|_{L^2(\Omega)}^2 \leqslant C_T \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right),$$

for a constant $C_T > 0$. Since the diffusion terms are nonnegative, we drop them and obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right) \leqslant C_T \left(\|z_1(t)\|_{L^2(\Omega)}^2 + \|z_2(t)\|_{L^2(\Omega)}^2 \right),$$

We use C_T for successive positive bounds. Since $z_1(0) = z_2(0) = 0$, Gronwall's lemma gives $z_1 = z_2 = 0$ in Q. The weak solution is therefore unique. □

Proposition 2.2 (Directional differentiability under a local stability assumption)

Fix $u_2 \in \mathscr{U}_{\mathrm{ad}}^\Omega$ and let $(v_1, v_2) = S(u_2) \in L^\infty(Q)^2$. We assume a local Lipschitz stability estimate for the control-to-state map. Thus, for some neighbourhood $\mathscr{V}$ of u_2 in $L^\infty(Q)$ and some $C > 0$, every $\widetilde{u}_2 \in \mathscr{V} \cap \mathscr{U}_{\mathrm{ad}}^\Omega$ has a state close to $S(u_2)$ and satisfies

$$\|S(\widetilde{u}_2) - S(u_2)\|_{L^\infty(Q)^2} \leqslant C \|\widetilde{u}_2 - u_2\|_{L^\infty(Q)}.$$

This estimate is standard under the boundedness assumptions used here [20].
Let $h \in L^\infty(Q)$ be admissible, so there is $\varepsilon_0 > 0$ such that

$$u_2 + \varepsilon h \in \mathscr{U}_{\mathrm{ad}}^\Omega \quad \text{for all } 0 \leqslant \varepsilon < \varepsilon_0.$$

Then the control-to-state map

$$S : \mathscr{U}_{\mathrm{ad}}^\Omega \longrightarrow L^2(Q)^2, \qquad u_2 \longmapsto (v_1, v_2)$$

is directionally differentiable at u_2 along h. The derivative $S'(u_2)h = (z_1, z_2)$ is the unique weak solution of

$$\begin{cases} \partial_t z_1 - \mu_1 \Delta z_1 = (-m + \alpha v_2) z_1 + \alpha v_1 z_2, & \text{in } Q, \\ \partial_t z_2 - \mu_2 \Delta z_2 = -\beta v_2 z_1 + (r - \beta v_1 - q u_2) z_2 - q v_2 h, & \text{in } Q, \\ \dfrac{\partial z_1}{\partial n} = \dfrac{\partial z_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ z_1(\cdot,0) = 0, \qquad z_2(\cdot,0) = 0, & \text{in } \Omega. \end{cases} \tag{5}$$

Equivalently,

$$\frac{S(u_2 + \varepsilon h) - S(u_2)}{\varepsilon} \underset{\varepsilon \to 0^+}{\longrightarrow} S'(u_2)h \quad \text{in } L^2(Q)^2.$$

Proof. For $0 < \varepsilon < \varepsilon_0$ small, local stability gives

$$\|S(u_2 + \varepsilon h) - S(u_2)\|_{L^\infty(Q)^2} \leqslant C|\varepsilon| \|h\|_{L^\infty(Q)}.$$

Set $u_2^\varepsilon = u_2 + \varepsilon h$ and $(v_1^\varepsilon, v_2^\varepsilon) = S(u_2^\varepsilon)$. Define the difference quotients

$$Z_i^\varepsilon := \frac{v_i^\varepsilon - v_i}{\varepsilon}, \qquad i = 1, 2.$$

Subtract the state system for (v_1, v_2) from the one for $(v_1^\varepsilon, v_2^\varepsilon)$ and divide by ε. Then

$$\begin{cases} \partial_t Z_1^\varepsilon - \mu_1 \Delta Z_1^\varepsilon = \dfrac{v_1^\varepsilon(-m + \alpha v_2^\varepsilon) - v_1(-m + \alpha v_2)}{\varepsilon}, \\ \partial_t Z_2^\varepsilon - \mu_2 \Delta Z_2^\varepsilon = \dfrac{v_2^\varepsilon(r - \beta v_1^\varepsilon) - q u_2^\varepsilon v_2^\varepsilon - v_2(r - \beta v_1) + q u_2 v_2}{\varepsilon}. \end{cases}$$

Using

$$v_i^\varepsilon = v_i + \varepsilon Z_i^\varepsilon, \qquad u_2^\varepsilon = u_2 + \varepsilon h,$$

the first quotient becomes

$$(-m + \alpha v_2) Z_1^\varepsilon + \alpha v_1 Z_2^\varepsilon + \varepsilon \alpha Z_1^\varepsilon Z_2^\varepsilon.$$

The second quotient is

$$-\beta v_2 Z_1^\varepsilon + (r - \beta v_1 - q u_2) Z_2^\varepsilon - q v_2 h - \varepsilon \beta Z_1^\varepsilon Z_2^\varepsilon - \varepsilon q h Z_2^\varepsilon.$$

Thus $(Z_1^\varepsilon, Z_2^\varepsilon)$ satisfies the linearized system (5), with additional quadratic remainders multiplied by ε. We first control the difference quotients uniformly. For small $\varepsilon > 0$, the states v_i^ε and v_i remain in a common bounded subset of $L^\infty(Q)$. The reaction maps are Lipschitz on this set. Applying the standard stability estimate for semilinear parabolic systems [6, 9, 20] to $v_i^\varepsilon - v_i$ gives

$$\|v_1^\varepsilon - v_1\|_{L^\infty(Q)} + \|v_2^\varepsilon - v_2\|_{L^\infty(Q)} \leqslant C|\varepsilon| \|h\|_{L^\infty(Q)}.$$

Thus,

$$\|Z_1^\varepsilon\|_{L^\infty(Q)} + \|Z_2^\varepsilon\|_{L^\infty(Q)} \leqslant C \|h\|_{L^\infty(Q)},$$

The constant C is independent of ε for $0 < \varepsilon < \varepsilon_0$ small. Thus $(Z_1^\varepsilon, Z_2^\varepsilon)$ is uniformly bounded in $L^\infty(Q)^2$, and hence also in $L^2(Q)^2$. The coefficients of the linearized system (5) belong to $L^\infty(Q)$, so standard linear parabolic theory gives a unique weak solution (z_1, z_2). We now compare this solution with the difference quotients. Set $R_i^\varepsilon = Z_i^\varepsilon - z_i$, $i = 1, 2$. The functions R_i^ε have zero initial data and homogeneous Neumann conditions. Subtracting (5) from the equations satisfied by Z_i^ε gives

$$\partial_t R_1^\varepsilon - \mu_1 \Delta R_1^\varepsilon = (-m + \alpha v_2) R_1^\varepsilon + \alpha v_1 R_2^\varepsilon + \varepsilon \alpha Z_1^\varepsilon Z_2^\varepsilon,$$

and

$$\partial_t R_2^\varepsilon - \mu_2 \Delta R_2^\varepsilon = -\beta v_2 R_1^\varepsilon + (r - \beta v_1 - q u_2) R_2^\varepsilon - \varepsilon \beta Z_1^\varepsilon Z_2^\varepsilon - \varepsilon q h Z_2^\varepsilon,$$

with homogeneous Neumann conditions and

$$R_1^\varepsilon(\cdot, 0) = R_2^\varepsilon(\cdot, 0) = 0.$$

The uniform L^∞ bounds for Z_1^ε and Z_2^ε imply

$$\|\varepsilon Z_1^\varepsilon Z_2^\varepsilon\|_{L^2(Q)} \leqslant C|\varepsilon| \|h\|_{L^\infty(Q)}^2, \quad \|\varepsilon h Z_2^\varepsilon\|_{L^2(Q)} \leqslant C|\varepsilon| \|h\|_{L^\infty(Q)}^2.$$

Test the remainder equations with R_1^ε and R_2^ε, integrate over Ω, and use the Neumann conditions. We obtain

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\|R_1^\varepsilon(t)\|_{L^2(\Omega)}^2+\|R_2^\varepsilon(t)\|_{L^2(\Omega)}^2\right)+\mu_1\|\nabla R_1^\varepsilon(t)\|_{L^2(\Omega)}^2+\mu_2\|\nabla R_2^\varepsilon(t)\|_{L^2(\Omega)}^2 \\ \leqslant C\left(\|R_1^\varepsilon(t)\|_{L^2(\Omega)}^2+\|R_2^\varepsilon(t)\|_{L^2(\Omega)}^2\right)+C\varepsilon^2.$$

Here $C>0$ depends on the state bounds, the model parameters, and $\|h\|_{L^\infty(Q)}$, but it is independent of ε. After removing the nonnegative diffusion terms and using the zero initial data, Gronwall's lemma gives

$$\|R_1^\varepsilon\|_{L^\infty(0,T;L^2(\Omega))}^2+\|R_2^\varepsilon\|_{L^\infty(0,T;L^2(\Omega))}^2\leqslant C\varepsilon^2.$$

Therefore $R_i^\varepsilon \xrightarrow[\varepsilon\to0^+]{} 0$ in $L^2(Q)$ and $Z_i^\varepsilon \xrightarrow[\varepsilon\to0^+]{} z_i$ in $L^2(Q)$ for $i=1,2$. This proves the directional differentiability of S along h and identifies $S'(u_2)h$. □

We now seek $u_2^*\in\mathscr{U}_{\mathrm{ad}}^\Omega$ such that

$$J(u_2^*)=\inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^\Omega}J(u_2),$$

under the constraint

$$\begin{cases}\partial_t v_1-\mu_1\Delta v_1=v_1(-m+\alpha v_2)\\ \partial_t v_2-\mu_2\Delta v_2=v_2(r-\beta v_1)-qu_2v_2\end{cases}$$

with nonnegative initial data $v_1(\cdot,0)=v_1^0$ and $v_2(\cdot,0)=v_2^0$ in Ω, together with $\partial_n v_1=\partial_n v_2=0$ on $\partial\Omega\times(0,T)$. We first prove that a minimizer exists.

Proposition 2.3 (Existence of an optimal control)

Let $\Omega\subseteq\mathbb{R}^d$ be bounded with sufficiently smooth boundary and set $Q=\Omega\times(0,T)$. Assume that all diffusion and reaction parameters, as well as $\gamma_1,\gamma_2,\lambda_2$, are positive, and let $\eta\geqslant0$. Suppose $v_1^0,v_2^0\in L^\infty(\Omega)$ are nonnegative a.e. and $v_1^{\mathrm{tar}},v_2^{\mathrm{tar}}\in L^2(Q)$.

Then there exists an admissible control

$$u_2^*\in\mathscr{U}_{\mathrm{ad}}^\Omega=\{u_2\in L^\infty(Q)\ \mid\ 0\leqslant u_2(x,t)\leqslant U_2^{\max}\ \text{a.e. in } Q\}$$

such that

$$J(u_2^*)=\inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^\Omega}J(u_2).$$

Proof. We use the direct method of the calculus of variations [12, 20]. Proposition 2.1 assigns to each $u_2\in\mathscr{U}_{\mathrm{ad}}^\Omega$ a unique nonnegative state $(v_1,v_2)\in L^\infty(Q)^2$. Since Q has finite measure and the targets belong to $L^2(Q)$, every tracking, effort, and yield term in

$$J(u_2)=\frac{1}{2}\int_Q\left[\gamma_1(v_1-v_1^{\mathrm{tar}})^2+\gamma_2(v_2-v_2^{\mathrm{tar}})^2+\lambda_2u_2^2\right]\mathrm{d}x\,\mathrm{d}t-\eta\int_Q qu_2v_2\,\mathrm{d}x\,\mathrm{d}t$$

is finite. We also need a lower bound. The tracking and effort terms are nonnegative, and only the yield term can be negative. Since $0\leqslant u_2\leqslant U_2^{\max}$ and

$$0\leqslant v_2(x,t)\leqslant M_2(T):=\|v_2^0\|_{L^\infty(\Omega)}e^{rT}\quad\text{a.e. in } Q,$$

we obtain

$$-\eta\int_Q qu_2v_2\,\mathrm{d}x\,\mathrm{d}t\geqslant-\eta qU_2^{\max}M_2(T)|Q|.$$

Consequently,

$$J(u_2)\geqslant-\eta qU_2^{\max}M_2(T)|Q|$$

for every $u_2\in\mathscr{U}_{\mathrm{ad}}^\Omega$. Choose a minimizing sequence $(u_2^k)_{k\geqslant1}\subset\mathscr{U}_{\mathrm{ad}}^\Omega$, so

$$J(u_2^k)\xrightarrow[k\to+\infty]{}\inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^\Omega}J(u_2).$$

The box constraint gives $\|u_2^k\|_{L^\infty(Q)}\leqslant U_2^{\max}$ for every k. By the Banach–Alaoglu theorem [6], the closed ball of $L^\infty(Q)$ is weakly-* compact. After extracting a subsequence, there is $u_2^*\in L^\infty(Q)$ such that

$$u_2^k\overset{*}{\underset{k\to+\infty}{\rightharpoonup}}u_2^*\quad\text{weakly-* in } L^\infty(Q).$$

The admissible set is weakly-* closed in $L^\infty(Q)$, so the limit u_2^* still satisfies $0 \leqslant u_2^* \leqslant U_2^{\max}$. Let $(v_1^k, v_2^k) = S(u_2^k)$. The comparison estimates from Proposition 2.1 are uniform in k because the same control bound holds for the whole sequence. Standard energy estimates for the two parabolic equations then give, for $i = 1, 2$,

$$(v_i^k)_{k\geqslant 1} \quad \text{bounded in } L^2(0,T;H^1(\Omega)), \quad \text{and} \quad (\partial_t v_i^k)_{k\geqslant 1} \quad \text{bounded in } L^2(0,T;V').$$

The reaction terms are uniformly bounded in $L^2(Q)$ because the controls and the two state components are uniformly bounded in $L^\infty(Q)$. The Aubin–Lions lemma [6, 20] therefore gives, after extraction of a subsequence, functions $\bar v_1, \bar v_2 \in L^2(Q)$ such that $v_i^k \xrightarrow[k\to+\infty]{} \bar v_i$ strongly in $L^2(Q)$ for $i = 1, 2$. The same uniform estimates give $v_i^k \xrightarrow[k\to+\infty]{*} \bar v_i$ weakly-* in $L^\infty(Q)$. Since every v_i^k is nonnegative, $\bar v_i \geqslant 0$ a.e. Strong L^2 convergence and the L^∞ bound also imply convergence in $L^p(Q)$ for every finite $p \geqslant 2$ along this subsequence. In particular, $v_1^k v_2^k \xrightarrow[k\to+\infty]{} \bar v_1 \bar v_2$ strongly in $L^1(Q)$, which is enough to pass to the bilinear reaction terms. For the control term,

$$u_2^k v_2^k \underset{k\to+\infty}{\rightharpoonup} u_2^* \bar v_2 \quad \text{in the sense of distributions.}$$

Indeed,

$$u_2^k v_2^k - u_2^* \bar v_2 = u_2^k (v_2^k - \bar v_2) + (u_2^k - u_2^*)\bar v_2.$$

The first term tends to zero in $L^1(Q)$ because (u_2^k) is bounded in $L^\infty(Q)$ and $v_2^k \xrightarrow[k\to+\infty]{} \bar v_2$ in $L^1(Q)$. For the second term, take $\varphi \in C_c^\infty(Q)$. Since $\bar v_2 \varphi \in L^1(Q)$,

$$\int_Q (u_2^k - u_2^*)\bar v_2 \varphi \,\mathrm{d}x\,\mathrm{d}t \xrightarrow[k\to+\infty]{} 0$$

by weak-* convergence in $L^\infty(Q)$. Hence $(\bar v_1, \bar v_2)$ solves the state system for u_2^*:

$$\begin{cases} \partial_t \bar v_1 - \mu_1 \Delta \bar v_1 = \bar v_1(-m + \alpha \bar v_2), & \text{in } Q, \\ \partial_t \bar v_2 - \mu_2 \Delta \bar v_2 = \bar v_2(r - \beta \bar v_1) - q u_2^* \bar v_2, & \text{in } Q, \\ \dfrac{\partial \bar v_1}{\partial n} = \dfrac{\partial \bar v_2}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ \bar v_1(\cdot,0) = v_1^0, \qquad \bar v_2(\cdot,0) = v_2^0, & \text{in } \Omega. \end{cases}$$

By uniqueness of the state equation, the limit pair is precisely $S(u_2^*) = (\bar v_1, \bar v_2)$. We now pass to the four terms of the cost. Since $v_i^k \xrightarrow[k\to+\infty]{} \bar v_i$ strongly in $L^2(Q)$, $i = 1, 2$,

$$\int_Q (v_i^k - v_i^{\mathrm{tar}})^2 \,\mathrm{d}x\,\mathrm{d}t \xrightarrow[k\to+\infty]{} \int_Q (\bar v_i - v_i^{\mathrm{tar}})^2 \,\mathrm{d}x\,\mathrm{d}t, \qquad i = 1, 2.$$

The sequence (u_2^k) is bounded in $L^\infty(Q)$ and therefore in $L^2(Q)$. Its weak-* limit in $L^\infty(Q)$ is also its weak limit in $L^2(Q)$. By weak lower semicontinuity of the squared L^2 norm,

$$\int_Q (u_2^*)^2 \,\mathrm{d}x\,\mathrm{d}t \leqslant \liminf_{k\to+\infty} \int_Q (u_2^k)^2 \,\mathrm{d}x\,\mathrm{d}t.$$

It remains to pass to the limit in the yield. Write

$$u_2^k v_2^k - u_2^* \bar v_2 = u_2^k (v_2^k - \bar v_2) + (u_2^k - u_2^*)\bar v_2.$$

The first term converges to zero in $L^1(Q)$ because $0 \leqslant u_2^k \leqslant U_2^{\max}$ and $v_2^k \xrightarrow[k\to+\infty]{} \bar v_2$ strongly in $L^1(Q)$. For the second term, the weak-* convergence of u_2^k can be tested against every function in $L^1(Q)$. Since $\bar v_2 \in L^1(Q)$, it follows that

$$\int_Q (u_2^k - u_2^*)\bar v_2 \,\mathrm{d}x\,\mathrm{d}t \xrightarrow[k\to+\infty]{} 0.$$

Therefore

$$\int_Q q u_2^k v_2^k \,\mathrm{d}x\,\mathrm{d}t \xrightarrow[k\to+\infty]{} \int_Q q u_2^* \bar v_2 \,\mathrm{d}x\,\mathrm{d}t.$$

Combining the three parts of the cost,

$$J(u_2^*) \leqslant \liminf_{k\to+\infty} J(u_2^k).$$

Since (u_2^k) is minimizing,

$$\lim_{k\to+\infty} J(u_2^k) = \inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

Hence

$$J(u_2^*) \leqslant \inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

Since $u_2^* \in \mathscr{U}_{\mathrm{ad}}^{\Omega}$, the reverse inequality also holds:

$$J(u_2^*) \geqslant \inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2).$$

We conclude that

$$J(u_2^*) = \inf_{u_2\in\mathscr{U}_{\mathrm{ad}}^{\Omega}} J(u_2),$$

so u_2^* is an optimal control. □

Remark. The compactness argument proves the existence of an optimal control [12, 20]. We now derive its first-order necessary conditions.

We keep the linearized state variables (z_1, z_2) separate from the adjoint variables (p_1, p_2), because they solve equations in opposite time directions and have different data. Set $y = (v_1, v_2)$. We place diffusion in the linear operator A and collect the local reactions and fishing term in F. The state system then has the compact form

$$\partial_t y = Ay + F(x,t,y,u_2), \qquad y = (v_1, v_2),$$

with

$$Ay = \begin{pmatrix} \mu_1\Delta v_1 \\ \mu_2\Delta v_2 \end{pmatrix}, \qquad F(x,t,v_1,v_2,u_2) = \begin{pmatrix} v_1(-m+\alpha v_2) \\ v_2(r-\beta v_1) - qu_2v_2 \end{pmatrix}.$$

Let L be the running cost and $\mathscr{L}$ the Lagrangian. The integrand of J is

$$L(x,t,v_1,v_2,u_2) = \frac{1}{2}\left[\gamma_1(v_1 - v_1^{\mathrm{tar}})^2 + \gamma_2(v_2 - v_2^{\mathrm{tar}})^2 + \lambda_2 u_2^2\right] - \eta q u_2 v_2.$$

For the adjoint $p = (p_1, p_2)$ [12, 19, 20], define the local Hamiltonian density by

$$\mathscr{H}(x,t,v_1,v_2,u_2,p_1,p_2) = L(x,t,v_1,v_2,u_2) + p\cdot F(x,t,v_1,v_2,u_2),$$

or, in full,

$$\begin{aligned}\mathscr{H}(x,t,v_1,v_2,u_2,p_1,p_2) := {}& \frac{1}{2}\left[\gamma_1(v_1 - v_1^{\mathrm{tar}})^2 + \gamma_2(v_2 - v_2^{\mathrm{tar}})^2 + \lambda_2 u_2^2\right] - \eta q u_2 v_2 \\ & + p_1 v_1(-m+\alpha v_2) + p_2\big(v_2(r-\beta v_1) - qu_2v_2\big).\end{aligned}$$

The local Hamiltonian contains the running cost, reactions, and control terms. We keep diffusion outside $\mathscr{H}$. The Lagrangian is

$$\begin{aligned}\mathscr{L}(v_1,v_2,u_2,p_1,p_2) := {}& \mathscr{J}(v_1,v_2,u_2) - \int_Q p_1\Big(\partial_t v_1 - \mu_1\Delta v_1 - v_1(-m+\alpha v_2)\Big)\,\mathrm{d}x\,\mathrm{d}t \\ & - \int_Q p_2\Big(\partial_t v_2 - \mu_2\Delta v_2 - v_2(r-\beta v_1) + qu_2v_2\Big)\,\mathrm{d}x\,\mathrm{d}t.\end{aligned}$$

To derive the adjoint equations from $\mathscr{L}$ [12, 20], take sufficiently regular variations δv_1 and δv_2. Since the initial state is fixed, $\delta v_i(\cdot,0) = 0$. Since the Neumann boundary condition is also fixed, the variations satisfy $\partial_n \delta v_i = 0$. We compute the Gâteaux derivative of $\mathscr{L}$ in each state direction:

$$\delta_{v_1}\mathscr{L}(\delta v_1) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathscr{L}(v_1+\varepsilon\delta v_1, v_2, u_2, p_1, p_2)\Big|_{\varepsilon=0}, \qquad \delta_{v_2}\mathscr{L}(\delta v_2) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathscr{L}(v_1, v_2+\varepsilon\delta v_2, u_2, p_1, p_2)\Big|_{\varepsilon=0}.$$

For the variation in v_1, the relevant terms of $\mathscr{L}$ are

$$\frac{1}{2}\int_Q \gamma_1(v_1 - v_1^{\mathrm{tar}})^2\,\mathrm{d}x\,\mathrm{d}t, \qquad -\int_Q p_1\big(\partial_t v_1 - \mu_1\Delta v_1 - v_1(-m+\alpha v_2)\big)\,\mathrm{d}x\,\mathrm{d}t,$$

and

$$-\int_Q p_2\big(\partial_t v_2 - \mu_2\Delta v_2 - v_2(r-\beta v_1) + qu_2v_2\big)\,\mathrm{d}x\,\mathrm{d}t.$$

Differentiation with respect to v_1 gives

$$\delta_{v_1}\mathscr{L}(\delta v_1) = \int_Q \gamma_1(v_1 - v_1^{\mathrm{tar}})\,\delta v_1 \,\mathrm{d}x\,\mathrm{d}t - \int_Q p_1\,\partial_t(\delta v_1)\,\mathrm{d}x\,\mathrm{d}t + \int_Q p_1\,\mu_1\Delta(\delta v_1)\,\mathrm{d}x\,\mathrm{d}t \\ + \int_Q p_1(-m + \alpha v_2)\,\delta v_1\,\mathrm{d}x\,\mathrm{d}t - \int_Q \beta p_2 v_2\,\delta v_1\,\mathrm{d}x\,\mathrm{d}t.$$

We integrate the derivative terms by parts. In time,

$$\int_Q p_1\,\partial_t(\delta v_1)\,\mathrm{d}x\,\mathrm{d}t = \left[\int_\Omega p_1(x,t)\,\delta v_1(x,t)\,\mathrm{d}x\right]_{t=0}^{T} - \int_Q \partial_t p_1\,\delta v_1\,\mathrm{d}x\,\mathrm{d}t.$$

Since $\delta v_1(\cdot,0) = 0$, only the terminal term remains:

$$\int_Q p_1\,\partial_t(\delta v_1)\,\mathrm{d}x\,\mathrm{d}t = \int_\Omega p_1(x,T)\,\delta v_1(x,T)\,\mathrm{d}x - \int_Q \partial_t p_1\,\delta v_1\,\mathrm{d}x\,\mathrm{d}t.$$

For diffusion, Green's identity gives

$$\int_\Omega p_1\Delta(\delta v_1)\,\mathrm{d}x = \int_\Omega \Delta p_1\,\delta v_1\,\mathrm{d}x + \int_{\partial\Omega}\left(p_1\frac{\partial \delta v_1}{\partial n} - \delta v_1\frac{\partial p_1}{\partial n}\right)\mathrm{d}S.$$

The state variations satisfy $\partial_n \delta v_1 = 0$ because the Neumann condition is fixed. We impose the same homogeneous condition $\partial_n p_1 = 0$ on the adjoint. The two boundary terms in Green's identity then vanish separately, and the diffusion contribution reduces to

$$\int_Q p_1\mu_1\Delta(\delta v_1)\,\mathrm{d}x\,\mathrm{d}t = \int_Q \mu_1\Delta p_1\,\delta v_1\,\mathrm{d}x\,\mathrm{d}t.$$

Collecting all terms,

$$\delta_{v_1}\mathscr{L}(\delta v_1) = \int_Q\Big[\gamma_1(v_1 - v_1^{\mathrm{tar}}) + \partial_t p_1 + \mu_1\Delta p_1 + p_1(-m + \alpha v_2) - \beta p_2 v_2\Big]\delta v_1\,\mathrm{d}x\,\mathrm{d}t - \int_\Omega p_1(x,T)\,\delta v_1(x,T)\,\mathrm{d}x.$$

Because δv_1 and $\delta v_1(\cdot,T)$ are arbitrary, their coefficients vanish. Thus $p_1(\cdot,T) = 0$ and

$$-\partial_t p_1 - \mu_1\Delta p_1 = \gamma_1(v_1 - v_1^{\mathrm{tar}}) + p_1(-m + \alpha v_2) - \beta p_2 v_2.$$

The same calculation for v_2 gives

$$\delta_{v_2}\mathscr{L}(\delta v_2) = \int_Q \gamma_2(v_2 - v_2^{\mathrm{tar}})\,\delta v_2\,\mathrm{d}x\,\mathrm{d}t - \int_Q p_2\,\partial_t(\delta v_2)\,\mathrm{d}x\,\mathrm{d}t + \int_Q p_2\,\mu_2\Delta(\delta v_2)\,\mathrm{d}x\,\mathrm{d}t \\ + \int_Q p_2\big(r - \beta v_1 - q\,u_2\big)\delta v_2\,\mathrm{d}x\,\mathrm{d}t + \int_Q \alpha p_1 v_1\,\delta v_2\,\mathrm{d}x\,\mathrm{d}t - \int_Q \eta q\,u_2\,\delta v_2\,\mathrm{d}x\,\mathrm{d}t.$$

After integration by parts, with $\partial_n \delta v_2 = 0$ and $\partial_n p_2 = 0$ on $\partial\Omega \times (0,T)$, we obtain

$$\delta_{v_2}\mathscr{L}(\delta v_2) = \int_Q\Big[\gamma_2(v_2 - v_2^{\mathrm{tar}}) + \partial_t p_2 + \mu_2\Delta p_2 - \eta q\,u_2 + \alpha p_1 v_1 + p_2(r - \beta v_1 - q\,u_2)\Big]\delta v_2\,\mathrm{d}x\,\mathrm{d}t \\ - \int_\Omega p_2(x,T)\,\delta v_2(x,T)\,\mathrm{d}x.$$

The arbitrary terminal variation gives $p_2(\cdot,T) = 0$, and the coefficient of δv_2 gives

$$-\partial_t p_2 - \mu_2\Delta p_2 = \gamma_2(v_2 - v_2^{\mathrm{tar}}) - \eta q\,u_2 + \alpha p_1 v_1 + p_2(r - \beta v_1 - q\,u_2).$$

The complete adjoint system is therefore

$$\begin{cases} -\partial_t p_1 - \mu_1\Delta p_1 = \gamma_1(v_1 - v_1^{\mathrm{tar}}) + p_1(-m + \alpha v_2) - \beta p_2 v_2, & \text{in } Q,\\ -\partial_t p_2 - \mu_2\Delta p_2 = \gamma_2(v_2 - v_2^{\mathrm{tar}}) - \eta q\,u_2 + \alpha p_1 v_1 + p_2(r - \beta v_1 - q\,u_2), & \text{in } Q,\\ \dfrac{\partial p_1}{\partial n} = \dfrac{\partial p_2}{\partial n} = 0, & \text{on } \partial\Omega\times(0,T),\\ p_1(\cdot,T) = 0, \qquad p_2(\cdot,T) = 0, & \text{in } \Omega. \end{cases}$$

The terminal conditions $p_1(\cdot,T) = p_2(\cdot,T) = 0$ come from the absence of a terminal cost: the values $v_i(\cdot,T)$ are free, so their terminal variations are arbitrary. The adjoint boundary conditions $\partial_n p_1 = \partial_n p_2 = 0$ cancel the boundary integrals created by Green's formula and match the homogeneous Neumann structure of the state equations.

Let h be an admissible direction and $(z_1, z_2) = S'(u_2)h$. Direct differentiation of the reduced cost gives

$$J'(u_2)h = \int_Q \gamma_1(v_1 - v_1^{\mathrm{tar}})z_1 + \left[\gamma_2(v_2 - v_2^{\mathrm{tar}}) - \eta q u_2\right] z_2 \,\mathrm{d}x\,\mathrm{d}t + \int_Q (\lambda_2 u_2 - \eta q v_2) h \,\mathrm{d}x\,\mathrm{d}t.$$

Multiply the linearized state equations by p_1, p_2 and integrate by parts in space and time. The terms involving z_1, z_2 cancel by the adjoint equations. Using

$$z_1(\cdot, 0) = z_2(\cdot, 0) = 0, \qquad p_1(\cdot, T) = p_2(\cdot, T) = 0,$$

and the homogeneous Neumann conditions, we obtain

$$J'(u_2)h = \int_Q \left[\lambda_2 u_2 - \eta q v_2 - q v_2 p_2\right] h \,\mathrm{d}x\,\mathrm{d}t.$$

Thus

$$J'(u_2)h = \int_Q \left[\lambda_2 u_2 - q v_2(\eta + p_2)\right] h \,\mathrm{d}x\,\mathrm{d}t.$$

The relations above are first-order necessary conditions. Under the box constraint, we take the derivative along admissible directions. Since $\mathscr{U}_{\mathrm{ad}}^{\Omega}$ is convex, every segment $u_2^* + s(w - u_2^*)$, $s \in [0,1]$, remains admissible. Let u_2^* be an optimal control, let $(v_1^*, v_2^*) = S(u_2^*)$, and let (p_1^*, p_2^*) solve the corresponding adjoint system. Differentiating along this segment at $s = 0$ gives the variational inequality

$$J'(u_2^*)(w - u_2^*) \geqslant 0 \qquad \text{for all } w \in \mathscr{U}_{\mathrm{ad}}^{\Omega}.$$

Using $\nabla J(u_2^*) = \lambda_2 u_2^* - q v_2^*(\eta + p_2^*)$, this is

$$\int_Q \left[\lambda_2 u_2^* - q v_2^*(\eta + p_2^*)\right](w - u_2^*) \,\mathrm{d}x\,\mathrm{d}t \geqslant 0 \qquad \text{for all } w \in \mathscr{U}_{\mathrm{ad}}^{\Omega}.$$

Since $\mathscr{U}_{\mathrm{ad}}^{\Omega}$ is defined pointwise by

$$0 \leqslant u_2(x,t) \leqslant U_2^{\max} \quad \text{for a.e. } (x,t) \in Q$$

the variational inequality is equivalent, a.e. in Q, to

$$u_2^*(x,t) = \Pi_{[0, U_2^{\max}]}\left(\frac{q\, v_2^*(x,t)}{\lambda_2}\big(\eta + p_2^*(x,t)\big)\right) \quad \text{for a.e. } (x,t) \in Q.$$

Here $\Pi_{[a,b]}(z) = \min(b, \max(a, z))$. Pointwise in (x,t), the second derivative of the Hamiltonian with respect to the control is $\partial^2_{u_2 u_2}\mathscr{H} = \lambda_2 > 0$. Hence $u_2 \mapsto \mathscr{H}$ is strictly convex and has a unique unconstrained minimizer, namely

$$\frac{q\, v_2^*(x,t)}{\lambda_2}\big(\eta + p_2^*(x,t)\big).$$

The box constraint $0 \leqslant u_2 \leqslant U_2^{\max}$ gives the projected formula above.

2.3 Stationary Profiles

We now study stationary profiles, namely time-independent solutions of the controlled spatial system. Such profiles are standard objects in reaction–diffusion theory [18]. They also serve as time-independent targets in population control [17]. A stationary profile is a pair (V_1, V_2) depending only on x. Substitution into the parabolic equations gives $\partial_t V_1 = \partial_t V_2 = 0$.

If the control is independent of time, write $u_2(x,t) = U_2(x)$. The stationary equations are

$$\begin{cases} -\mu_1 \Delta V_1 = V_1(-m + \alpha V_2), & \text{in } \Omega, \\ -\mu_2 \Delta V_2 = V_2(r - \beta V_1) - q U_2 V_2, & \text{in } \Omega, \\ \dfrac{\partial V_1}{\partial n} = \dfrac{\partial V_2}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

We consider both constant and spatially varying stationary controls U_2. In either case, (V_1, V_2) solves a coupled nonlinear elliptic system [6, 18]. The product V_1V_2 makes this elliptic system nonlinear. Homogeneous Neumann conditions admit constant functions because $\partial_n V_i = 0$ automatically. Therefore every equilibrium of the non-spatial controlled system gives a constant stationary profile. When U_2 is constant, the diffusion terms vanish for such a profile and the stationary equations reduce to

$$\begin{cases} V_1(-m + \alpha V_2) = 0, \\ V_2(r - \beta V_1 - qU_2) = 0. \end{cases}$$

The positive constant profile is

$$V_1^* = \frac{r - qU_2}{\beta}, \qquad V_2^* = \frac{m}{\alpha},$$

provided $r - qU_2 > 0$, or equivalently $qU_2 < r$. If $r - qU_2 \leqslant 0$, the positive homogeneous coexistence profile is absent. Nonconstant stationary profiles satisfy the same Neumann conditions and have $\nabla V_i \not\equiv 0$. We write this nonlinear stationary problem as the operator equation

$$F(V_1, V_2) = 0 \quad \text{where} \quad F(V_1, V_2) = \begin{pmatrix} F_1(V_1, V_2) \\ F_2(V_1, V_2) \end{pmatrix},$$

with

$$F_1(V_1, V_2) = -\mu_1 \Delta V_1 - V_1(-m + \alpha V_2), \qquad F_2(V_1, V_2) = -\mu_2 \Delta V_2 - V_2(r - \beta V_1 - qU_2).$$

After expanding the reaction terms,

$$F_1(V_1, V_2) = -\mu_1 \Delta V_1 + mV_1 - \alpha V_1 V_2, \qquad F_2(V_1, V_2) = -\mu_2 \Delta V_2 - rV_2 + \beta V_1 V_2 + qU_2 V_2.$$

The bilinear product V_1V_2 makes the elliptic operator F nonlinear. Newton's method solves $F(V_1, V_2) = 0$ by successive linearizations [11]. Starting from an initial pair $(V_1^{(0)}, V_2^{(0)})$, each iteration computes a correction by solving the linearized equation

$$DF(V_1^{(k)}, V_2^{(k)}) \begin{pmatrix} \delta V_1^{(k)} \\ \delta V_2^{(k)} \end{pmatrix} = -F(V_1^{(k)}, V_2^{(k)}),$$

and updating

$$V_1^{(k+1)} = V_1^{(k)} + \delta V_1^{(k)}, \qquad V_2^{(k+1)} = V_2^{(k)} + \delta V_2^{(k)}.$$

The Fréchet derivative of the first component is

$$\delta F_1 = -\mu_1 \Delta(\delta V_1) + m\delta V_1 - \alpha V_2 \delta V_1 - \alpha V_1 \delta V_2 = -\mu_1 \Delta(\delta V_1) + (m - \alpha V_2)\delta V_1 - \alpha V_1 \delta V_2.$$

For the second component,

$$\delta F_2 = -\mu_2 \Delta(\delta V_2) + \beta V_2 \delta V_1 + (-r + \beta V_1 + qU_2)\delta V_2.$$

At iteration k, the Newton correction $(\delta V_1^{(k)}, \delta V_2^{(k)})$ therefore solves

$$\begin{cases} -\mu_1 \Delta \delta V_1^{(k)} + \big(m - \alpha V_2^{(k)}\big)\delta V_1^{(k)} - \alpha V_1^{(k)} \delta V_2^{(k)} = -F_1(V_1^{(k)}, V_2^{(k)}), & \text{in } \Omega, \\ \beta V_2^{(k)} \delta V_1^{(k)} - \mu_2 \Delta \delta V_2^{(k)} + \big(-r + \beta V_1^{(k)} + qU_2\big)\delta V_2^{(k)} = -F_2(V_1^{(k)}, V_2^{(k)}), & \text{in } \Omega, \\ \dfrac{\partial \delta V_1^{(k)}}{\partial n} = \dfrac{\partial \delta V_2^{(k)}}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

The equations are linear in the corrections $\delta V_1^{(k)}$ and $\delta V_2^{(k)}$, but their coefficients are evaluated at the current iterate. After finite-difference discretization, multiplication by $V_i^{(k)}$ becomes a diagonal matrix and diffusion becomes the Neumann matrix Δ_h. If the two grid vectors are stacked into one vector, the Newton equation takes the block form

$$\begin{pmatrix} -\mu_1 \Delta_h + mI - \alpha \operatorname{diag}(V_2^{(k)}) & -\alpha \operatorname{diag}(V_1^{(k)}) \\ \beta \operatorname{diag}(V_2^{(k)}) & -\mu_2 \Delta_h - rI + \beta \operatorname{diag}(V_1^{(k)}) + q \operatorname{diag}(U_2) \end{pmatrix} \begin{pmatrix} \delta V_1^{(k)} \\ \delta V_2^{(k)} \end{pmatrix} = - \begin{pmatrix} F_1(V_1^{(k)}, V_2^{(k)}) \\ F_2(V_1^{(k)}, V_2^{(k)}) \end{pmatrix}.$$

The matrix Δ_h is the same reflected Neumann Laplacian used in the reaction–diffusion simulation. Its assembly and the sparsity pattern of the Newton matrix can be reused. The coefficients of the full Newton matrix depend on the

current iterate, so its numerical factorization is recomputed at each Newton step. If U_2 is constant and $r - qU_2 > 0$, the positive homogeneous coexistence profile provides the natural Newton initial guess

$$(V_1^*, V_2^*) = \left(\frac{r - qU_2}{\beta}, \frac{m}{\alpha} \right),$$

To test the local behavior of this constant state, we can add a small Neumann-compatible perturbation:

$$V_1^{(0)}(x) = V_1^* + \varepsilon\phi(x), \quad V_2^{(0)}(x) = V_2^* + \varepsilon\psi(x),$$

Here $\varepsilon > 0$ is small and the perturbations satisfy $\partial_n\phi = \partial_n\psi = 0$. We study the local stationary equations analytically near the homogeneous state. For constant U_2, homogeneous coefficients, and Neumann boundary conditions, decompose a perturbation into Laplacian eigenmodes. The mode-by-mode linearization below shows that no small-amplitude nonconstant stationary branch bifurcates from the positive coexistence profile. We therefore linearize the stationary system at

$$V_1^* = \frac{r - qU_2}{\beta}, \quad V_2^* = \frac{m}{\alpha}.$$

Expand the perturbations in Neumann eigenfunctions [6, 18]. Let

$$-\Delta\phi_k = \lambda_k\phi_k, \qquad \frac{\partial\phi_k}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

The mode $\lambda_0 = 0$ is constant. Every $\lambda_k > 0$ is spatially varying. For a fixed mode, set

$$V_1 = V_1^* + \xi_1\phi_k, \quad V_2 = V_2^* + \xi_2\phi_k.$$

The linearized coefficients satisfy

$$\begin{pmatrix} \mu_1\lambda_k & -\alpha V_1^* \\ \beta V_2^* & \mu_2\lambda_k \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 0.$$

A nonzero vector (ξ_1, ξ_2) exists only when the determinant is zero:

$$\mu_1\mu_2\lambda_k^2 + \alpha\beta V_1^* V_2^* = 0.$$

In the coexistence regime, $V_1^*, V_2^* > 0$ and $\mu_1, \mu_2, \alpha, \beta > 0$. Hence

$$\mu_1\mu_2\lambda_k^2 + \alpha\beta V_1^* V_2^* > 0$$

for every Neumann mode k. The determinant is strictly positive for $k = 0$ and for all $k \geqslant 1$, so the linearized stationary operator has no kernel on any eigenspace of the Neumann Laplacian. The linearized stationary equations therefore have only the zero solution in every Neumann mode. We now choose a spatially varying control $U_2(x)$ to obtain nonconstant numerical profiles. We solve

$$\begin{cases} -\mu_1\Delta V_1 = V_1(-m + \alpha V_2), & \text{in } \Omega, \\ -\mu_2\Delta V_2 = V_2(r - \beta V_1) - qU_2(x)V_2, & \text{in } \Omega, \\ \dfrac{\partial V_1}{\partial n} = \dfrac{\partial V_2}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

For Figure 8, the code [2] solves the stationary equations directly with `scipy.optimize.least_squares`. It minimizes the squared Euclidean norm of the finite-difference residual, with bounds $10^{-8} \leqslant V_{i,j} \leqslant 5$ and solver tolerances 10^{-11}. We use 120 points on $[0, 1]$, so $h = 1/119$, and the reflected Neumann stencil. The parameters are $m = 0.6$, $r = 1.2$, $\alpha = \beta = q = 1$, and $\mu_1 = \mu_2 = 10^{-3}$. The prescribed control is

$$U_2(x) = 0.4 + 0.9\exp\left(-\frac{(x - 0.5)^2}{2(0.12)^2} \right).$$

A rerun gives a maximum absolute stationary residual of about 1.6×10^{-11}. The resulting densities are positive and nonconstant: the spatially varying fishing effort changes the local reaction balance.

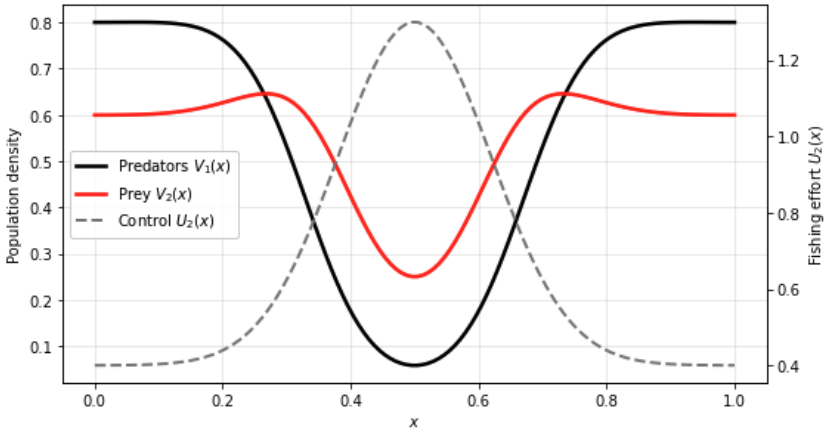


Figure 8: Nonconstant stationary densities generated by a spatially heterogeneous control $U_2(x)$.

2.4 Numerical Computation of the Optimal Control

We now discretize and solve the optimality system. The unknowns are the states (v_1, v_2), the adjoints (p_1, p_2), and the control u_2. A forward–backward sweep performs, at each iteration, one forward state solve, one backward adjoint solve, and one pointwise control update. Forward–backward sweeps are classical for optimality systems derived from Pontryagin's principle, and their convergence has been studied in [13]. We first present a semi-implicit first-order version in which diffusion is solved implicitly and the local terms remain explicit. This gives a transparent and inexpensive reference algorithm. We then explain the stability and convergence limits of this fixed-point iteration and compare it with two reduced optimization methods, projected gradient and L-BFGS-B. In all three cases, special attention is given to the discrete adjoint: it is built from the fully discrete state recurrence so that the computed vector is the gradient of the discrete cost actually minimized by the code.

A first-order semi-implicit forward–backward sweep method

We use a first-order semi-implicit time discretization [10, 16]. Diffusion is evaluated at the new time level, while the nonlinear reactions and the control term are evaluated at the old time level. Each forward step therefore requires two linear elliptic solves. This choice removes the diffusion restriction of a fully explicit method and keeps the implementation simple. We follow a discretize–then–optimize strategy [12, 20]. First, the state recurrence and the quadrature defining the cost are fixed. Next, the discrete adjoint is obtained by transposing the linearized discrete state map. The resulting vector is exactly the gradient of the finite-dimensional reduced objective. Let $t_n = n\Delta t$, $n = 0, \dots, N$, with $\Delta t = T/N$. We approximate u_2 by a control that is constant on each interval $[t_n, t_{n+1})$ and denote its value by $u_{2,n}$, $n = 0, \dots, N-1$. Let Δ_h be the finite-difference Laplacian with homogeneous Neumann conditions. On the uniform grid, it is self-adjoint for the discrete L^2 inner product used in the cost. Consequently, the adjoint diffusion solve uses the adjoint of $I - \Delta t\, \mu_i \Delta_h$, which has the same form as the forward matrix in this inner product.

The complete first-order forward–backward sweep is as follows.

Input: final time T, number of time steps N, time step $\Delta t = T/N$, domain $\Omega \subseteq \mathbf{R}^d$, spatial mesh, initial states $v_{1,0} = v_1^0$ and $v_{2,0} = v_2^0$, control bound $U_2^{\max}$, model and cost parameters, target profiles, relaxation $0 < \theta \leqslant 1$, tolerance $\varepsilon > 0$, and maximum iteration count $K_{\max}$.

Initialization:

- set $t_n = n\Delta t$, $n = 0, \dots, N$,
- choose the spatial grid on Ω,
- choose $u_{2,n}^{(0)}(x) \in [0, U_2^{\max}]$, $n = 0, \dots, N-1$, for example $u_{2,n}^{(0)} \equiv 0$ or a constant,
- set $k = 0$ and $E^{(0)} = +\infty$.

While $k < K_{\max}$ and $E^{(k)} \geqslant \varepsilon$:

- **Step 1: forward solution of the state system.**

  Starting from $v_{1,0}^{(k)} = v_1^0$ and $v_{2,0}^{(k)} = v_2^0$, compute $n = 0, \dots, N-1$:

$$\begin{cases} (I - \Delta t\, \mu_1 \Delta_h)\, v_{1,n+1}^{(k)} = v_{1,n}^{(k)} + \Delta t\, v_{1,n}^{(k)}\big(-m + \alpha v_{2,n}^{(k)}\big), \\ (I - \Delta t\, \mu_2 \Delta_h)\, v_{2,n+1}^{(k)} = v_{2,n}^{(k)} + \Delta t \left[v_{2,n}^{(k)}\big(r - \beta v_{1,n}^{(k)}\big) - q\, u_{2,n}^{(k)} v_{2,n}^{(k)} \right]. \end{cases}$$

  Each forward step therefore consists of two linear elliptic solves.

- **Step 2: backward solution of the discrete adjoint system.**

  The discrete cost is evaluated at t_n, $n = 0, \dots, N-1$. Since there is no terminal term, set $p_{1,N}^{(k)}(x) = p_{2,N}^{(k)}(x) = 0$.

  For $n = N-1, \dots, 0$, solve

$$\begin{cases} \dfrac{p_{1,n}^{(k)}(x) - p_{1,n+1}^{(k)}(x)}{\Delta t} - \mu_1 \Delta_h p_{1,n}^{(k)}(x) = \gamma_1 \big(v_{1,n}^{(k)}(x) - v_{1,n}^{\mathrm{tar}}(x)\big) + \big(-m + \alpha v_{2,n}^{(k)}(x)\big) p_{1,n+1}^{(k)}(x) \\ \qquad\qquad - \beta v_{2,n}^{(k)}(x) p_{2,n+1}^{(k)}(x), & \text{in } \Omega, \\ \dfrac{p_{2,n}^{(k)}(x) - p_{2,n+1}^{(k)}(x)}{\Delta t} - \mu_2 \Delta_h p_{2,n}^{(k)}(x) = \gamma_2 \big(v_{2,n}^{(k)}(x) - v_{2,n}^{\mathrm{tar}}(x)\big) - \eta q\, u_{2,n}^{(k)}(x) + \alpha v_{1,n}^{(k)}(x) p_{1,n+1}^{(k)}(x) \\ \qquad\qquad + \big(r - \beta v_{1,n}^{(k)}(x) - q\, u_{2,n}^{(k)}(x)\big) p_{2,n+1}^{(k)}(x), & \text{in } \Omega, \\ \dfrac{\partial p_{1,n}^{(k)}}{\partial n} = \dfrac{\partial p_{2,n}^{(k)}}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

  Step 3: control update.

  For $n = 0, \dots, N-1$ and at every spatial node, compute the unprojected control

$$u_{r,2,n}^{(k+1)}(x) = \frac{q\, v_{2,n}^{(k)}(x)}{\lambda_2} \left(\eta + p_{2,n+1}^{(k)}(x) \right).$$

  Then apply relaxation and projection:

$$u_{2,n}^{(k+1)}(x) = \Pi_{[0, U_2^{\max}]} \left(\theta u_{r,2,n}^{(k+1)}(x) + (1-\theta) u_{2,n}^{(k)}(x) \right), \qquad 0 < \theta \leqslant 1.$$

  The parameter $\theta \in (0, 1]$ mixes the new pointwise value with the previous iterate. Its numerical effect is studied below. The projection is defined by

$$\Pi_{[0, U_2^{\max}]}(z) = \min\left(U_2^{\max}, \max(0, z)\right).$$

  Thus $u_{2,n}^{(k+1)}(x) \in [0, U_2^{\max}]$ at every mesh point.

  Step 4: convergence test.

  Compute

$$E^{(k+1)} = \max_{0 \leqslant n \leqslant N-1} \left\| u_{2,n}^{(k+1)} - u_{2,n}^{(k)} \right\|_{L^\infty(\Omega)}.$$

  On the grid, $E^{(k+1)}$ is computed with the discrete maximum norm.

  If $E^{(k+1)} < \varepsilon$, stop.

  Otherwise set $k \leftarrow k+1$ and repeat.

After the last control update, recompute the state and adjoint with the final control.

Output: discrete state and adjoint trajectories $v_{1,n}^{\mathrm{opt}}, v_{2,n}^{\mathrm{opt}}, p_{1,n}^{\mathrm{opt}}, p_{2,n}^{\mathrm{opt}}$, $n = 0, \dots, N$, and the discrete control $u_{2,n}^{\mathrm{opt}}$, $n = 0, \dots, N-1$.

Time accuracy and optimization methods

The method has order one in time: diffusion uses implicit Euler and the local terms use explicit Euler. The explicit reaction terms determine the choice of time step. The forward–backward sweep is a fixed-point iteration. The contraction bound $K_h < 1$ in the next subsection ensures convergence, and a fixed point satisfies the discrete optimality system. We also use projected-gradient and limited-memory quasi-Newton methods to decrease the discrete cost directly [4, 5, 22].

Reduced formulation and projected-gradient method

A second strategy treats the same fully discrete problem as a finite-dimensional optimization problem and minimizes the reduced cost directly. Let $S_h : u_h \mapsto v_h$ denote the discrete control-to-state map, where

$$u_h := \big(u_{2,n}(x_i)\big)_{\substack{0\leqslant n\leqslant N-1\\ 0\leqslant i\leqslant M}} \in \mathbf{R}^{N(M+1)}$$

collects the $N(M+1)$ control values and $v_h = S_h(u_h)$ is the full state trajectory. Define

$$j_h(u_h) := J_h(S_h(u_h), u_h).$$

The finite-dimensional problem is therefore $\min_{u_h\in\mathscr{U}_{\mathrm{ad},h}} j_h(u_h)$, where

$$\mathscr{U}_{\mathrm{ad},h} := \left\{ u_h = \big(u_{2,n}(x_i)\big)_{\substack{0\leqslant n\leqslant N-1\\ 0\leqslant i\leqslant M}} \in \mathbf{R}^{N(M+1)} \;\middle|\; 0 \leqslant u_{2,n}(x_i) \leqslant U_2^{\max} \text{ for all } n, i \right\}.$$

The reduced formulation removes the state from the list of optimization variables. One forward solve and one backward adjoint solve provide all components of the gradient. For the one-dimensional benchmark, set $A_i = I - \Delta t\,\mu_i\Delta_h$. The backward sweep stores $p_{i,n} = h^{-1}(A_i^T)^{-1}\ell_{i,n}$, where $\ell_{i,n}$ is the canonical discrete adjoint of j_h. This change of variables gives the stated backward recurrence exactly. With the time-space inner product $\Delta t\,h\sum_{n,j} a_{n,j}b_{n,j}$, the gradient component at time t_n is

$$[\nabla j_h(u_h)]_n = \lambda_2 u_{2,n} - \eta q v_{2,n} - q v_{2,n} p_{2,n+1}.$$

The code uses the Euclidean gradient

$$\nabla_E j_h(u_h) = \Delta t\,h\,\nabla j_h(u_h).$$

We use this Euclidean gradient in all optimization updates and reported residuals below. Define

$$R_h(u_h) = u_h - \Pi_{\mathscr{U}_{\mathrm{ad},h}}\big(u_h - \nabla_E j_h(u_h)\big).$$

Then $R_h(u_h) = 0$ is equivalent to the box-constrained first-order optimality condition. Following the projected-gradient method [4], set $g_h^{(k)} = \nabla_E j_h(u_h^{(k)})$ and update

$$u_h^{(k+1)} = \Pi_{\mathscr{U}_{\mathrm{ad},h}}\left(u_h^{(k)} - \rho_k g_h^{(k)}\right).$$

The projection is componentwise, so every iterate satisfies the control bounds. The benchmark code [2] starts each line search with $\rho_k = 1$ and tests the displacement

$$d_k = \Pi_{\mathscr{U}_{\mathrm{ad},h}}\left(u_h^{(k)} - \rho_k g_h^{(k)}\right) - u_h^{(k)}.$$

Its sufficient-decrease test is

$$j_h(u_h^{(k)} + d_k) \leqslant j_h(u_h^{(k)}) - 10^{-4}\|d_k\|_2^2,$$

where $\|d_k\|_2^2$ is the unweighted sum of the squared control components. If the test fails, the code halves ρ_k. It performs at most 20 tests. If all tests fail, it takes the projected step with $\rho_k = 2^{-20}$ without another decrease test. A step accepted by the displayed test reduces the discrete cost when $d_k \neq 0$.

Projected gradient stops before an update if $\|R_h(u_h^{(k)})\|_\infty < 10^{-6}$, or after an update if $\|u_h^{(k+1)} - u_h^{(k)}\|_\infty < 10^{-6}$. In this implementation, each line-search trial calls the routine that evaluates both the cost and the gradient. Each trial therefore performs an additional forward state solve and an additional backward adjoint solve, although its gradient is discarded.

Quasi-Newton reduced optimization

A third option is the bound-constrained quasi-Newton method L-BFGS-B [5, 22]. After discretization, the distributed control becomes a vector $u_h \in \mathbf{R}^{N(M+1)}$ with componentwise constraints $0 \leqslant u_h \leqslant U_2^{\max}$. The state variables are still eliminated through S_h, so the only optimization unknown is u_h and the reduced objective is

$$j_h(u_h) = J_h(S_h(u_h), u_h).$$

The value $j_h(u_h)$ is obtained from one forward state solve, and $\nabla_E j_h(u_h)$ is obtained from one backward discrete adjoint solve. L-BFGS-B then builds a limited-memory approximation of the inverse reduced Hessian. More precisely, it stores only a small number of differences $s_k = u_h^{(k+1)} - u_h^{(k)}$ and $y_k = \nabla_E j_h(u_h^{(k+1)}) - \nabla_E j_h(u_h^{(k)})$. These pairs contain local curvature information and define a quasi-Newton search direction without forming the full Hessian in $\mathbf{R}^{N(M+1)\times N(M+1)}$. The method also identifies active components at the lower and upper bounds.

Comparison of the methods

The three methods use the same state equation and the same adjoint-consistent gradient. FBS is a low-cost fixed-point baseline. Projected gradient tests cost decrease with a line search. L-BFGS-B is a box-constrained quasi-Newton method that also uses curvature information.

Numerical comparison of the optimization methods

For the numerical benchmark [2], we take $\Omega = (0, 1)$ and $T = 4$. The benchmark uses $M_x = 80$ spatial points, so $h = 1/(M_x - 1)$. It uses $N = 120$ time steps. In Python, the parameter M counts points. In the indexing $i = 0, \ldots, M$ above, M counts intervals, so it equals $M_x - 1$. We set $m = 1$, $r = 1.2$, $\alpha = \beta = q = 1$, and $\mu_1 = \mu_2 = 0.002$. Homogeneous Neumann conditions are used at both endpoints. The localized initial densities are

$$v_1^0(x) = 1.3 \exp\left(-\frac{(x - 0.30)^2}{2 \cdot 0.10^2}\right) + 0.05, \quad v_2^0(x) = 1.4 \exp\left(-\frac{(x - 0.70)^2}{2 \cdot 0.10^2}\right) + 0.05.$$

The targets are time-independent perturbations of the uncontrolled coexistence state $(r/\beta, m/\alpha)$:

$$v_1^{\mathrm{tar}}(x) = \frac{r}{\beta} + 0.15 \cos(2\pi x), \qquad v_2^{\mathrm{tar}}(x) = \frac{m}{\alpha} - 0.15 \cos(2\pi x).$$

The admissible set is defined by $0 \leqslant u_2(x, t) \leqslant U_2^{\max}$ with $U_2^{\max} = 2$. We choose $\gamma_1 = \gamma_2 = 0.5$, $\lambda_2 = 0.05$, and $\eta = 0.8$. Every algorithm starts from the same admissible control $u_2^{(0)} \equiv 0$ and uses at most 100 outer iterations. FBS stops when the maximum control change is below 10^{-6}. Its default relaxation is $\theta = 0.5$. Projected gradient uses the line search and the two stopping tests described above. L-BFGS-B uses the same control bounds, a projected-gradient tolerance of 10^{-6} (`gtol`), a relative cost tolerance of 10^{-12} (`ftol`), and a memory of 10 curvature pairs (`maxcor`). The spatial grid, the time grid, the state solver, the cost quadrature, and the discrete adjoint are identical in all three tests. The comparison therefore concerns the optimization updates, not different discretizations. We report the final value of $j_h(u_h)$, the iteration count, the CPU time, and the projected-gradient residual

$$\left\| u_h - \Pi_{\mathscr{U}_{\mathrm{ad},h}}\left(u_h - \nabla_E j_h(u_h)\right) \right\|_\infty .$$

This residual measures the violation of the projected first-order condition under the box constraints. It is zero at a discrete stationary point and remains meaningful when some components of the control lie on 0 or $U_2^{\max}$. The discrete objective contains the reward term $-\eta q u_2 v_2$, so a negative final value means that the weighted yield exceeds the positive tracking and effort contributions. Table 1 compares FBS with $\theta = 0.5$, projected gradient with Armijo backtracking, and reduced L-BFGS-B.

Method	$j_h(u_h)$	Iterations	CPU time (s)	Residual
FBS, $\theta = 0.5$	2.696596	100	0.688705	6.91×10^{-3}
Projected gradient	1.913330	100	1.423093	3.68×10^{-3}
L-BFGS-B	-1.552726	38	0.616277	8.53×10^{-7}

Table 1: Comparison of the optimization methods.

L-BFGS-B gives the lowest cost and residual and stops after the fewest iterations. Since the FBS result depends on θ, we also use

$$\theta \in \{1, 0.75, 0.5, 0.25, 0.1, 0.05\}.$$

The parameter θ sets the weight of the new control before projection. We compare the six values listed above using the same initial control and 100 FBS sweeps. For each sweep, we record the RMS norm of the control update, the cost, and the mean control. The plots use the Python code in the GitHub repository [2].

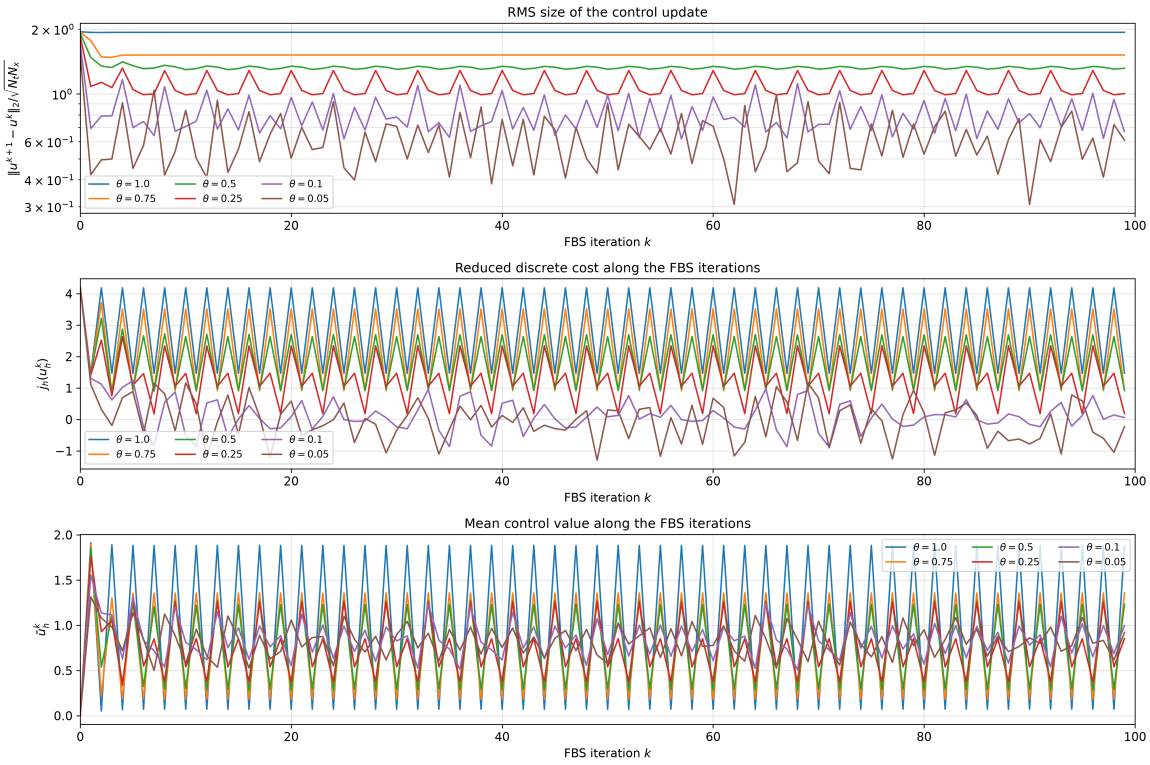


Figure 9: Influence of θ on the forward–backward sweep. The panels show the RMS norm of $u_h^{k+1} - u_h^k$, the cost $j_h(u_h^k)$, and the mean control. Smaller values of θ reduce the size of the updates. All six runs still oscillate after 100 sweeps.

Figure 9 shows persistent oscillations for all six values of θ. The RMS update stays far above the stopping tolerance 10^{-6}. None of the runs reaches this tolerance within 100 sweeps. For $\theta = 1$, the cost alternates between about 1.470647 and 4.189759. The plots show $k = 0, \ldots, 99$: the cost and mean control are evaluated at u_h^k, and the update is $u_h^{k+1} - u_h^k$. Table 2 reports the values after the last update, at u_h^{100}. The horizontal axis counts FBS iterations k. The script records one value after each sweep. We checked the effect of the mesh size with an additional NumPy implementation of the repository's FBS formulas. We ran 100 sweeps with $\theta = 1$. We used 80 or 160 spatial points and 120 or 240 time steps, giving four meshes. Over the last 20 sweeps, the cost ranges from 1.470647 to 4.189759 on the original mesh. It ranges from 1.451476 to 3.967468 when both meshes are refined. The nearly alternating behavior remains in all four tests. In Table 2, $\overline{u}_h$ denotes the average of the control over all spatial nodes and time intervals.

θ	$j_h(u_h)$	Iterations	CPU time (s)	Residual	$\overline{u}_h$
1.00	4.189759	100	0.700941	6.811×10^{-3}	0.072166
0.75	3.517737	100	0.690913	6.820×10^{-3}	0.183496
0.50	2.696596	100	0.688705	6.908×10^{-3}	0.286550
0.25	2.345476	100	0.689804	7.265×10^{-3}	0.377919
0.10	0.439558	100	0.689668	5.857×10^{-3}	0.650697
0.05	-0.330494	100	0.695693	9.18×10^{-4}	0.941267

Table 2: Influence of the relaxation parameter θ on the forward–backward sweep method. The cost, residual, and mean control are from the regenerated runs. CPU times are from the reference benchmark.

Among the tested values, $\theta = 0.05$ gives the best FBS iterate after 100 sweeps: it has the smallest FBS cost and the smallest FBS residual in Table 2. It still remains above the L-BFGS-B result, which reaches a lower cost and a residual below 10^{-6} after 38 iterations. Projected gradient also reaches the prescribed iteration limit. For this parameter set, L-BFGS-B is the most effective solver in terms of cost, stationarity, and iteration count, while the

forward–backward sweep remains useful as a simple reference implementation.

Interpretation in terms of fishing.

At a control satisfying the projected first-order condition, $u_2(x,t)$ describes the fishing effort. At points where $u_2 = U_2^{\max}$, the marginal gain $qv_2(\eta + p_2)$ is at least the marginal effort cost $\lambda_2 U_2^{\max}$. The upper bound is therefore active.

2.5 Fully Discrete Conditional Convergence of the Forward–Backward Sweep Method

We finish with a conditional convergence theorem for the fully discrete forward–backward sweep, following the contraction estimates in [13]. The spatial mesh, the time step, and the final time are fixed in this section. We prove convergence of the resulting finite-dimensional map. Let Δ_h be the discrete Laplacian with homogeneous Neumann conditions. We assume that Δ_h is nonpositive and self-adjoint for the discrete inner product used to define the adjoint. The benchmark code uses a symmetric Neumann matrix. Its interior rows are $(w_{j-1} - 2w_j + w_{j+1})/h^2$. The two boundary rows are $(w_1 - w_0)/h^2$ and $(w_{M_x-2} - w_{M_x-1})/h^2$. The matrix is nonpositive and self-adjoint for $\langle a, b\rangle_h = h\sum_{j=0}^{M_x-1} a_j b_j$. This inner product is h times the Euclidean inner product. The earlier simulations used the reflected ghost-point stencil. With $\mu_i \geqslant 0$, $i = 1, 2$, define

$$A_i := I - \Delta t\, \mu_i \Delta_h.$$

Let A_i^* denote the adjoint of A_i for this discrete inner product. We assume that both A_i and A_i^* are invertible and satisfy the maximum-norm stability bounds

$$\|A_i^{-1}\|_\infty \leqslant 1, \qquad \|(A_i^*)^{-1}\|_\infty \leqslant 1, \qquad i = 1, 2.$$

These estimates hold, for example, for M-matrices with a discrete maximum principle [10, 15, 16]. We can check these bounds directly for the benchmark matrix. The off-diagonal entries of Δ_h are nonnegative, and each row sums to zero. Thus A_i has nonpositive off-diagonal entries. In each row, its diagonal entry exceeds the sum of the absolute values of its other entries by 1. Hence A_i is an invertible M-matrix, and A_i^{-1} has nonnegative entries. Also, $A_i\mathbf{1} = \mathbf{1}$, so $A_i^{-1}\mathbf{1} = \mathbf{1}$. The rows of A_i^{-1} therefore sum to 1. Since its entries are nonnegative, $\|A_i^{-1}\|_\infty = 1$. Finally, symmetry gives $A_i^* = A_i^T = A_i$. The same bound holds for $(A_i^*)^{-1}$. For $w_h = (w_n)_{n=0}^N$, set

$$\|w_h\|_{\infty,h} := \max_{0\leqslant n\leqslant N} \|w_n\|_{\ell^\infty}.$$

For $u_{2,h} = (u_{2,n})_{n=0}^{N-1}$, set

$$\|u_{2,h}\|_{\infty,h} := \max_{0\leqslant n\leqslant N-1} \|u_{2,n}\|_{\ell^\infty}.$$

For $w_h = (w_{1,h}, w_{2,h})$, use the product norm

$$\|w_h\|_{\infty,h} := \max_{0\leqslant n\leqslant N} \left(\|w_{1,n}\|_{\ell^\infty} + \|w_{2,n}\|_{\ell^\infty}\right).$$

The discrete admissible set is

$$\mathscr{U}_{\mathrm{ad},h} = \{u_{2,h} = (u_{2,n})_{0\leqslant n\leqslant N-1} \ : \ 0 \leqslant u_{2,n}(x_j) \leqslant U_2^{\max} \text{ for all } n \text{ and all mesh points } x_j\}.$$

The finite-dimensional set $\mathscr{U}_{\mathrm{ad},h}$ is closed, hence complete. For $u_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$, the state recursion is

$$\begin{cases} A_1 v_{1,n+1} = v_{1,n} + \Delta t\, v_{1,n}(-m + \alpha v_{2,n}), \\ A_2 v_{2,n+1} = v_{2,n} + \Delta t\, [v_{2,n}(r - \beta v_{1,n}) - q u_{2,n} v_{2,n}]. \end{cases}$$

The adjoint is then computed backward from the zero terminal data $p_{1,N} = p_{2,N} = 0$. For $n = N-1, \dots, 0$, its recursion is

$$\begin{cases} A_1^* p_{1,n} = p_{1,n+1} + \Delta t\, G_{1,n}, \\ A_2^* p_{2,n} = p_{2,n+1} + \Delta t\, G_{2,n}. \end{cases}$$

where

$$G_{1,n} := \gamma_1(v_{1,n} - v_{1,n}^{\mathrm{tar}}) + p_{1,n+1}(-m + \alpha v_{2,n}) - \beta p_{2,n+1} v_{2,n},$$

and

$$G_{2,n} := \gamma_2(v_{2,n} - v_{2,n}^{\mathrm{tar}}) - \eta q\, u_{2,n} + \alpha p_{1,n+1} v_{1,n} + p_{2,n+1}(r - \beta v_{1,n} - q u_{2,n}).$$

The benchmark matrix is symmetric, so $A_i^* = A_i^T = A_i$. We can therefore use the same matrix factorization for the forward and backward diffusion solves. Using the projected optimality condition from the previous subsection, define the relaxed control map by

$$\Phi_h(u_{2,h})_n = \Pi_{[0,U_2^{\max}]}\left((1-\theta)u_{2,n} + \theta\frac{qv_{2,n}}{\lambda_2}(\eta + p_{2,n+1})\right), \qquad 0 < \theta \leqslant 1,$$

for $n = 0, \dots, N-1$. The projection guarantees $\Phi_h(u_{2,h}) \in \mathscr{U}_{\mathrm{ad},h}$, so Φ_h is a self-map of the complete admissible set. The relaxed FBS iteration is $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$. If $u_{2,h} = \Phi_h(u_{2,h})$, then the relaxation term cancels and the same projected optimality condition is obtained for every $\theta \in (0,1]$.

Theorem 2.4 (Fully discrete conditional convergence)

Assume that the state and adjoint recursions, with the matrices A_i^*, are defined for every $u_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$. Let $0 < \theta \leqslant 1$, and suppose that all generated trajectories satisfy the uniform bounds

$$\|v_{1,h}\|_{\infty,h} + \|v_{2,h}\|_{\infty,h} \leqslant M_v, \qquad \|p_{1,h}\|_{\infty,h} + \|p_{2,h}\|_{\infty,h} \leqslant M_p,$$

Assume also that the discrete reactions and the adjoint source terms are Lipschitz on the bounded set determined by M_v, M_p, and $U_2^{\max}$. This assumption is automatic here because every local term is a polynomial in the bounded variables. The forward recurrence then depends Lipschitz-continuously on the control, and the backward recurrence depends Lipschitz-continuously on the state and control. Thus there are constants $C_{S,h}, C_{P,h} > 0$, independent of the particular pair $u_{2,h}, \widehat{u}_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$, such that

$$\|v_h - \widehat{v}_h\|_{\infty,h} \leqslant C_{S,h}\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}, \quad \|p_h - \widehat{p}_h\|_{\infty,h} \leqslant C_{P,h}\left(\|v_h - \widehat{v}_h\|_{\infty,h} + \|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}\right),$$

where $v_h, \widehat{v}_h$ are generated by $u_{2,h}, \widehat{u}_{2,h}$, and $p_h, \widehat{p}_h$ are their adjoints. The constants $C_{S,h}$ and $C_{P,h}$ are chosen for the fixed spatial discretization, Δt, the final time T, the biological and cost parameters, and the bounds M_v, M_p. They are uniform with respect to the two admissible controls being compared. Define

$$K_h := \frac{q}{\lambda_2}\left[(\eta + M_p)C_{S,h} + M_v C_{P,h}(1 + C_{S,h})\right].$$

Then

$$\|\Phi_h(u_{2,h}) - \Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \leqslant \kappa_h\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}, \qquad \kappa_h := (1-\theta) + \theta K_h.$$

Since $0 < \theta \leqslant 1$, $\kappa_h < 1$ if and only if $K_h < 1$. Under this condition, Φ_h is a contraction on $\mathscr{U}_{\mathrm{ad},h}$. Banach's fixed-point theorem then gives a unique fixed point $u_{2,h}^{\infty} \in \mathscr{U}_{\mathrm{ad},h}$. For every initial control $u_{2,h}^{(0)}$, the iterates $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$ converge to $u_{2,h}^{\infty}$ in the discrete maximum norm. The Lipschitz estimates for the forward and backward recursions then give convergence of the associated states and adjoints. Passing to the limit in all three updates shows that the limiting quintuple satisfies the fully discrete optimality system.

Proof. Choose two controls $u_{2,h}, \widehat{u}_{2,h} \in \mathscr{U}_{\mathrm{ad},h}$. Let $(v_{1,h}, v_{2,h})$ and $(\widehat{v}_{1,h}, \widehat{v}_{2,h})$ be the associated state trajectories, and let $(p_{1,h}, p_{2,h})$ and $(\widehat{p}_{1,h}, \widehat{p}_{2,h})$ be their discrete adjoints. For $n = 0, \dots, N$, set

$$\delta v_{i,n} := v_{i,n} - \widehat{v}_{i,n}, \qquad \delta p_{i,n} := p_{i,n} - \widehat{p}_{i,n}, \qquad i = 1, 2.$$

For $n = 0, \dots, N-1$, set $\delta u_{2,n} := u_{2,n} - \widehat{u}_{2,n}$.

Step 1: Lipschitz dependence of the discrete state on the control.

Subtracting the two state recursions gives

$$\begin{cases} A_1\delta v_{1,n+1} = \delta v_{1,n} + \Delta t\left[v_{1,n}(-m + \alpha v_{2,n}) - \widehat{v}_{1,n}(-m + \alpha\widehat{v}_{2,n})\right], \\ A_2\delta v_{2,n+1} = \delta v_{2,n} + \Delta t\left[R_2(v_{1,n}, v_{2,n}, u_{2,n}) - R_2(\widehat{v}_{1,n}, \widehat{v}_{2,n}, \widehat{u}_{2,n})\right], \end{cases}$$

where $R_2(v_1, v_2, u_2) := v_2(r - \beta v_1) - q u_2 v_2$.

On the bounded set fixed by M_v and $U_2^{\max}$, the reaction terms are Lipschitz. Hence, for some $L_S > 0$,

$$\|\delta v_{1,n+1}\|_{\ell^\infty} + \|\delta v_{2,n+1}\|_{\ell^\infty} \leqslant (1 + \Delta t L_S)\left(\|\delta v_{1,n}\|_{\ell^\infty} + \|\delta v_{2,n}\|_{\ell^\infty}\right) + \Delta t\, qM_v\|\delta u_{2,n}\|_{\ell^\infty}.$$

Using $\|A_i^{-1}\|_\infty \leqslant 1$, define

$$E_n^v := \|\delta v_{1,n}\|_{\ell^\infty} + \|\delta v_{2,n}\|_{\ell^\infty}.$$

Since both states have the same initial data, $E_0^v = 0$, and

$$E_{n+1}^v \leqslant (1 + \Delta t L_S)E_n^v + \Delta t\, qM_v\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}.$$

The discrete Gronwall lemma [10, 16] gives

$$E_n^v \leqslant qM_vTe^{L_ST}\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}.$$

Therefore

$$\|v_h - \widehat{v}_h\|_{\infty,h} \leqslant C_{S,h}\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h},$$

with, for example, $C_{S,h} = qM_vTe^{L_ST}$.

Step 2: Lipschitz dependence of the discrete adjoint on the state and control.

Both adjoint trajectories start from the same terminal value: $p_{1,N} = \widehat{p}_{1,N} = 0$ and $p_{2,N} = \widehat{p}_{2,N} = 0$. Subtracting the two backward recursions gives

$$\begin{cases} A_1^*\delta p_{1,n} = \delta p_{1,n+1} + \Delta t\, \delta G_{1,n}, \\ A_2^*\delta p_{2,n} = \delta p_{2,n+1} + \Delta t\, \delta G_{2,n}. \end{cases}$$

On the bounded set fixed by M_v and M_p, the maps G_1, G_2 are Lipschitz. Thus there are $L_P > 0$ and $C_P^0 > 0$ such that

$$\|\delta G_{1,n}\|_{\ell^\infty} + \|\delta G_{2,n}\|_{\ell^\infty} \leqslant L_P\left(\|\delta p_{1,n+1}\|_{\ell^\infty} + \|\delta p_{2,n+1}\|_{\ell^\infty}\right) + C_P^0\left(\|\delta v_{1,n}\|_{\ell^\infty} + \|\delta v_{2,n}\|_{\ell^\infty} + \|\delta u_{2,n}\|_{\ell^\infty}\right).$$

Using $\|(A_i^*)^{-1}\|_\infty \leqslant 1$ and

$$E_n^p := \|\delta p_{1,n}\|_{\ell^\infty} + \|\delta p_{2,n}\|_{\ell^\infty},$$

we obtain

$$E_n^p \leqslant (1 + \Delta t L_P)E_{n+1}^p + \Delta t\, C_P^0\left(\|v_h - \widehat{v}_h\|_{\infty,h} + \|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}\right).$$

Since $E_N^p = 0$, backward discrete Gronwall gives

$$\|p_h - \widehat{p}_h\|_{\infty,h} \leqslant C_{P,h}\left(\|v_h - \widehat{v}_h\|_{\infty,h} + \|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}\right),$$

where $C_{P,h} = C_P^0 Te^{L_PT}$ is admissible. With the state estimate,

$$\|p_h - \widehat{p}_h\|_{\infty,h} \leqslant C_{P,h}(1 + C_{S,h})\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}.$$

Step 3: Lipschitz estimate for the control update.

In this step, $v_{2,h}$ and $\widehat{v}_{2,h}$ denote their restrictions to the control times $n = 0, \dots, N-1$. The projection $\Pi_{[0,U_2^{\max}]}$ is 1-Lipschitz. Therefore

$$\|\Phi_h(u_{2,h}) - \Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \leqslant (1-\theta)\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h} + \theta\frac{q}{\lambda_2}\left\|v_{2,h}(\eta + p_{2,h}^+) - \widehat{v}_{2,h}(\eta + \widehat{p}_{2,h}^+)\right\|_{\infty,h},$$

where

$$p_{2,h}^+ := (p_{2,n+1})_{0\leqslant n\leqslant N-1}, \qquad \widehat{p}_{2,h}^+ := (\widehat{p}_{2,n+1})_{0\leqslant n\leqslant N-1}.$$

Split the product as

$$v_{2,h}(\eta + p_{2,h}^+) - \widehat{v}_{2,h}(\eta + \widehat{p}_{2,h}^+) = (v_{2,h} - \widehat{v}_{2,h})(\eta + p_{2,h}^+) + \widehat{v}_{2,h}(p_{2,h}^+ - \widehat{p}_{2,h}^+).$$

The bounds $\|v_{2,h}\|_{\infty,h} \leqslant M_v$ and $\|p_{2,h}^+\|_{\infty,h} \leqslant M_p$ give

$$\left\|v_{2,h}(\eta + p_{2,h}^+) - \widehat{v}_{2,h}(\eta + \widehat{p}_{2,h}^+)\right\|_{\infty,h} \leqslant (\eta + M_p)\|v_{2,h} - \widehat{v}_{2,h}\|_{\infty,h} + M_v\|p_{2,h}^+ - \widehat{p}_{2,h}^+\|_{\infty,h}.$$

Moreover,

$$\|p_{2,h}^+ - \widehat{p}_{2,h}^+\|_{\infty,h} \leqslant \|p_h - \widehat{p}_h\|_{\infty,h},$$

so the estimates from Steps 1 and 2 yield

$$\|\Phi_h(u_{2,h}) - \Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \leqslant \left[(1-\theta) + \theta\frac{q}{\lambda_2}\left((\eta + M_p)C_{S,h} + M_vC_{P,h}(1 + C_{S,h})\right)\right]\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}.$$

By the definition of K_h and κ_h,

$$\|\Phi_h(u_{2,h}) - \Phi_h(\widehat{u}_{2,h})\|_{\infty,h} \leqslant \kappa_h\|u_{2,h} - \widehat{u}_{2,h}\|_{\infty,h}, \quad \text{with} \quad \kappa_h := (1-\theta) + \theta K_h.$$

Step 4: conclusion.

The preceding estimate proves that Φ_h is Lipschitz with constant $\kappa_h = (1-\theta) + \theta K_h$. Since $0 < \theta \leqslant 1$, we have $\kappa_h < 1$ exactly when $K_h < 1$. Under this condition, Φ_h maps the complete metric space $\mathscr{U}_{\mathrm{ad},h}$ into itself and is a contraction. Banach's fixed-point theorem gives a unique fixed point $u_{2,h}^{\infty}$. For every initial admissible control, the sequence $u_{2,h}^{(k+1)} = \Phi_h(u_{2,h}^{(k)})$ converges to this point in $\|\cdot\|_{\infty,h}$, with the geometric estimate $\|u_{2,h}^{(k)} - u_{2,h}^{\infty}\|_{\infty,h} \leqslant \kappa_h^k \|u_{2,h}^{(0)} - u_{2,h}^{\infty}\|_{\infty,h}$. The state and adjoint Lipschitz estimates then imply

$$v_{i,h}^{(k)} \xrightarrow[k\to+\infty]{} v_{i,h}^{\infty} \quad \text{and} \quad p_{i,h}^{(k)} \xrightarrow[k\to+\infty]{} p_{i,h}^{\infty}, \qquad i = 1, 2.$$

Passing to the limit in the state recursion, the adjoint recursion, and the projected control formula shows that

$$(v_{1,h}^{\infty}, v_{2,h}^{\infty}, p_{1,h}^{\infty}, p_{2,h}^{\infty}, u_{2,h}^{\infty})$$

solves the fully discrete optimality system, which completes the proof. □

Remark. For fixed a priori and Lipschitz constants, K_h contains the ratio q/λ_2. Thus $K_h < 1$ is easier to satisfy when q is small or λ_2 is large. This relation also has a modeling meaning: q is the fishing efficiency and λ_2 is the cost of effort. Small q means low fishing efficiency, while large λ_2 means expensive effort.

Conclusion

This report studied the spatial Lotka–Volterra model and the control of prey harvesting. Finite differences describe population movement, and a sequential time update gives nonnegative numerical solutions under the stated step-size bounds. The simulations show how the populations evolve with Neumann and Dirichlet boundary conditions. A spatially varying fishing effort also produces nonconstant stationary densities.

For the control problem, we proved existence of an optimal control and derived the state equation, the adjoint equation, and the projected first-order condition. The discrete adjoint provides the exact gradient of the discrete cost. In the reported benchmark, FBS keeps oscillating for all six tested relaxation parameters, whereas L-BFGS-B reaches a much smaller projected residual and a lower cost. The contraction theorem explains FBS convergence when $K_h < 1$. These results connect the analysis of the model with the behavior of the numerical algorithms.

References

[1] H. Amann. Existence and regularity for semilinear parabolic evolution equations. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 11 (4):593–676, 1984. URL https://numdam.org/item/ASNSP_1984_4_11_4_593_0/.

[2] A. Autran. Lotka–Volterra. https://github.com/AleXXandreAutran/Lotka-Volterra, 2026. GitHub repository.

[3] M. Beck and M. J. Gander. On the positivity of Poisson integrators for the Lotka–Volterra equations. *BIT Numerical Mathematics*, 55(2):319–340, 2015. doi: 10.1007/s10543-014-0505-1.

[4] D. P. Bertsekas. *Nonlinear Programming*. Athena Scientific, Belmont, MA, 2nd edition, 1999. ISBN 978-1-886529-00-7.

[5] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu. A limited-memory algorithm for bound-constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi: 10.1137/0916069.

[6] L. C. Evans. *Partial Differential Equations*, volume 19 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2nd edition, 2010. doi: 10.1090/gsm/019.

[7] M. J. Gander. A non spiraling integrator for the Lotka–Volterra equation. *Il Volterriano*, pages 21–28, 1994. URL https://archive-ouverte.unige.ch/unige:6309. No. 4.

[8] M. J. Gander and F. Kwok. *Numerical Analysis of Partial Differential Equations Using Maple and MATLAB*. Fundamentals of Algorithms. Society for Industrial and Applied Mathematics, Philadelphia, 2018. ISBN 978-1-61197-530-7. doi: 10.1137/1.9781611975314.

[9] D. Henry. *Geometric Theory of Semilinear Parabolic Equations*, volume 840 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, 1981.

[10] W. Hundsdorfer and J. G. Verwer. *Numerical Solution of Time-Dependent Advection–Diffusion–Reaction Equations*, volume 33 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2003.

[11] C. T. Kelley. *Iterative Methods for Linear and Nonlinear Equations*, volume 16 of *Frontiers in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, 1995.

[12] J.-L. Lions. *Optimal Control of Systems Governed by Partial Differential Equations*. Springer-Verlag, Berlin, 1971.

[13] M. McAsey, L. Mou, and W. Han. Convergence of the forward–backward sweep method in optimal control. *Computational Optimization and Applications*, 53(1):207–226, 2012. doi: 10.1007/s10589-011-9454-7.

[14] J. D. Murray. *Mathematical Biology I: An Introduction*, volume 17 of *Interdisciplinary Applied Mathematics*. Springer, New York, 3rd edition, 2002.

[15] A. Quarteroni and A. Valli. *Numerical Approximation of Partial Differential Equations*, volume 23 of *Springer Series in Computational Mathematics*. Springer-Verlag, Berlin, 1994.

[16] A. Quarteroni, R. Sacco, and F. Saleri. *Numerical Mathematics*, volume 37 of *Texts in Applied Mathematics*. Springer, Berlin, 2nd edition, 2007.

[17] Q. Richard. *Comportement asymptotique de modèles de populations structurées*. PhD thesis, Université Bourgogne Franche-Comté, 2018. URL https://hal.science/tel-01891708v2. NNT: 2018UBFCD050.

[18] J. Smoller. *Shock Waves and Reaction–Diffusion Equations*, volume 258 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, New York, 2nd edition, 1994.

[19] E. Trélat. Contrôle optimal, 2008. URL https://www.ljll.fr/trelat/enseignement/M2controle_optimal/courscontopt.pdf. Lecture notes, Master's program in Mathematics, Université d'Orléans, academic year 2007–2008.

[20] F. Tröltzsch. *Optimal Control of Partial Differential Equations: Theory, Methods and Applications*, volume 112 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2010.

[21] V. Volterra. Variazioni e fluttuazioni del numero d'individui in specie animali conviventi. *Memorie del R. Comitato Talassografico Italiano*, CXXXI, 1927.

[22] C. Zhu, R. H. Byrd, P. Lu, and J. Nocedal. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4):550–560, 1997. doi: 10.1145/279232.279236.